DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
SCHOOL OF ENGINEERING
MANIPAL UNIVERSITY JAIPUR

LAB MANUAL

MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

Course Name: Digital Electronics Lab

Course Code ECE2131

B.Tech. (ECE) II Year, III Semester

Contents

S.No.	Contents	Page no.
1.	Evaluation Scheme	2
2.	Do's and Don'ts	3
3.	Laboratory Instructions	4
4.	List of Experiments	5-6
5.	Introduction to Laboratory components	7-17

Evaluation Scheme:

Class: IIIrd Semester B.Tech.	Evaluation
Branch: ECE & VLSI Schedule per Week Practical Hrs : 2 hr/week	Examination Time = Three (3) Hours Maximum Marks = 100 [Internal (60) & External (40)]

DO'S AND DON'T'S

DO'S

- Students should get the record of previous experiment checked before starting the new experiment.
- Read the manual carefully before starting the experiment.
- Before starting the experiment, get circuit diagram checked by the teacher.
- Before switching on the power supply, get the circuit connections checked.
- Get your readings checked by the teacher.
- Apparatus must be handled carefully.
- Maintain strict discipline.
- Keep your mobile phone switched off or in vibration mode.
- Students should get the experiment allotted for next turn, before leaving the lab.

DON'TS

- Do not touch or attempt to touch the mains power supply wire with bare hands.
- Do not overcrowd the tables.
- Do not tamper with equipments.
- Do not leave the lab without prior permission from the teacher.

Instructions to the Students

General Instructions

- Maintain separate observation copy for each laboratory.
- Observations or readings should be taken only in the observation copy.
- Get the readings counter signed by the faculty after the completion of the experiment.
- Maintain Index column in the observation copy and get the signature of the faculty before leaving the lab.

Before Entering the Lab

- The previous experiment should have been written in the practical file, without which the students will not be allowed to enter the lab.
- The students should have written the experiment in the observation copy that they are supposed to perform in the lab.
- The experiment written in the observation copy should have aim, apparatus required, circuit diagram/algorithm, blank observation table (if any), formula (if any), programmed (if any), model graph (if any) and space for result.

When Working in the Lab

- Necessary equipment/apparatus should be taken only from the lab assistant by making an issuing slip, which would contain name of the experiment, names of batch members and apparatus or components required.
- Never switch on the power supply before getting the permission from the faculty.

Before Leaving the Lab

- The equipment/components should be returned back to the lab assistant in good condition after the completion of the experiment.
- The students should get the signature from the faculty in the observation copy.
- They should also check whether their file is checked and counter signed in the index.

List of Experiments

Exp. No.	Experiments	
1.	Study and Implementation of Logic Gates	
	i)	Verification of truth table for basic logic gates (AND, OR, NOT, NAND, NOR , XOR)
	ii)	Study of Universality of NAND and NOR gates and realization of basic gates
2.	Study of basic arithmetic circuits	
	i)	Design and study of Half adder/ Half Subtractor
	ii)	Design and study of Full adder/ Full Subtractor
3.	Design and implementation of combinational circuits using NAND and NOR gates	
	i)	Design, build and test simple combinational circuits using logic gates like NAND and NOR
4.	Code converters Circuits	
	i)	Design of 4-bit Binary to Gray Code converter
	ii)	Design of BCD to Excess-3 Code converter
5.	Magnitude Comparator and Parity Checker/Generator (Odd/even parity)	
	i)	Study and implementation of 1-bit magnitude comparator
	ii)	Design and study of 4-bit magnitude comparator using IC 7485
	iii)	Design and implementation of Parity Checker and Generator (for Odd/even parity)
6.	Study of Multiplexers	
	i)	Implement 4:1 multiplexer using IC 74153.
	ii)	Implement f = ∑ m (0, 3, 5, 7, 8, 10, 14) using i) 8:1 MUX
7	Study of Demultiplexer	
	i)	Design a 1:4 De-multiplexer using basic gates
	ii)	Design a 1:8 De-multiplexer using IC.
8.	Encoders and Decoders	
	i)	Design and implement 2 to 4 decoder using basic gates
	ii)	Design and implement 2 to 4 decoder with enable input using only NAND gates
9.	Basics of Flip-Flops	
	i)	SR Flip-flop using NAND gates

	ii)	JK Flip-Flop using NAND gates
	iii)	T-Flip Flop using NAND gates
10.	Study of Flip-Flop ICs	
	i)	Study of IC 7474: Positive edge triggered dual D flip-flops with preset and clear
	ii)	Study of 74LS73 A: Negative edge triggered dual JK Flip-flops with clear
11.	Study of Counters.	
	i)	Implement Synchronous 4-bit up counter using IC 7473
	ii)	Design 4 bit Ring Counter
	iii)	Design Johnson Counter
12.	Study of Shift Registers	
	i)	Design SISO, SIPO using D Flip-flop,
	ii)	Design PISO and PIPO register using D Flipflop.
13.	Design of Finite State Machine.	
	i)	Design a sequence detector which will detect an overlapping sequence 101 from an input sequence.
14.		Study of logic family
	i)	Implement TTL logic for NAND and NOR Gate.

Introduction to Laboratory components

The Breadboard

The breadboard consists of two terminal strips and two bus strips (often broken in the centre). Each bus strip has two rows of contacts. Each of the two rows of contacts are a node. That is, each contact along a row on a bus strip is connected together (inside the breadboard). Bus strips are used primarily for power supply connections, but are also used for any node requiring a large number of connections. Each terminal strip has 60 rows and 5 columns of contacts on each side of the centre gap. Each row of 5 contacts is a node. You will build your circuits on the terminal strips by inserting the leads of circuit components into the contact receptacles and making connections with 22-26 gauge wire. There are wire cutter/strippers and a spool of wire in the lab. It is a good practice to wire +5V and 0V power supply connections to separate bus strips.

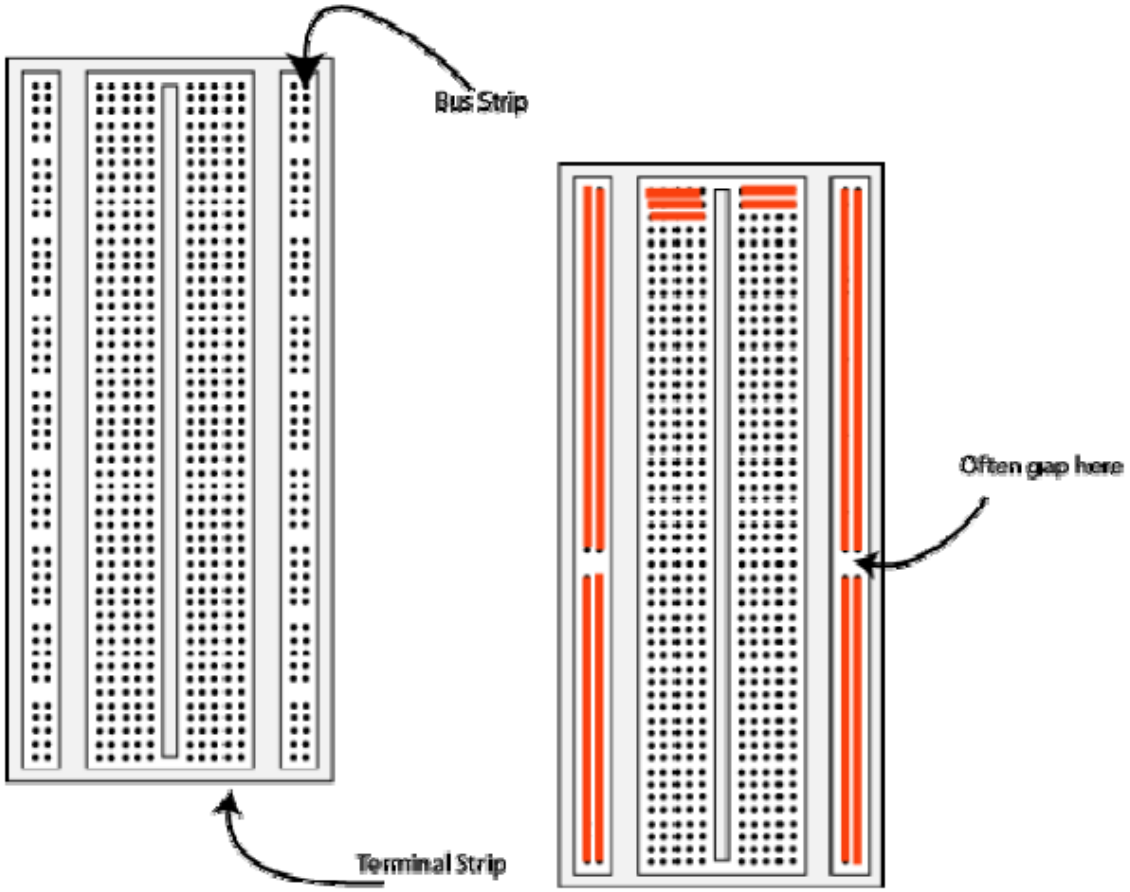


Fig. 1 : The breadboard-The lines indicate connected holes

The 5V supply MUST NOT BE EXCEEDED since this will damage the ICs (Integrated circuits) used during the experiments. Incorrect connection of power to the ICs could result in them exploding or becoming very hot - with the possible serious injury occurring to the people working on the experiment! Ensure that the power supply polarity and all components and connections are correct before switching on power .

Building the Circuit:

Throughout these experiments we will use TTL chips to build circuits. The steps for wiring a circuit should be completed in the order described below:

1. Turn the power (Trainer Kit) off before you build anything!
2. Make sure the power is off before you build anything!
3. Connect the +5V and ground (GND) leads of the power supply to the power and ground bus strips on your breadboard.
4. Plug the chips you will be using into the breadboard. Point all the chips in the same direction with pin 1 at the upper-left corner. (Pin 1 is often identified by a dot or a notch next to it on the chip package)
5. Connect +5V and GND pins of each chip to the power and ground bus strips on the breadboard.
6. Select a connection on your schematic and place a piece of hook-up wire between corresponding pins of the chips on your breadboard. It is better to make the short connections before the longer ones. Mark each connection on your schematic as you go, so as not to try to make the same connection again at a later stage.
7. Get one of your group members to check the connections, before you turn the power on.
8. If an error is made and is not spotted before you turn the power on. Turn the power off immediately before you begin to rewire the circuit.
9. At the end of the laboratory session, collect you hook-up wires, chips and all equipment and return them to the demonstrator.
10. Tidy the area that you were working in and leave it in the same condition as it was before you started.

Common Causes of Problems:

1. Not connecting the ground and/or power pins for all chips.
2. Not turning on the power supply before checking the operation of the circuit.
3. Leaving out wires.
4. Plugging wires into the wrong holes.
5. Driving a single gate input with the outputs of two or more gates
6. Modifying the circuit with the power on.

In all experiments, you will be expected to obtain all instruments, leads, components at the start of the experiment and return them to their proper place after you have finished the experiment. Please inform the demonstrator or technician if you locate faulty

equipment. If you damage a chip, inform a demonstrator, don't put it back in the box of chips for somebody else to use.

Pin diagram of various logic gates:

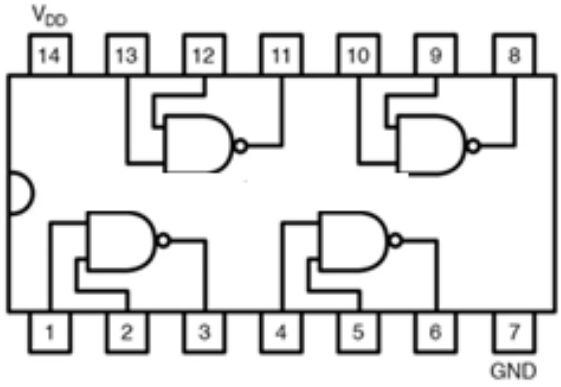


Fig. 2: Quad 2 Input NAND (IC-7400)

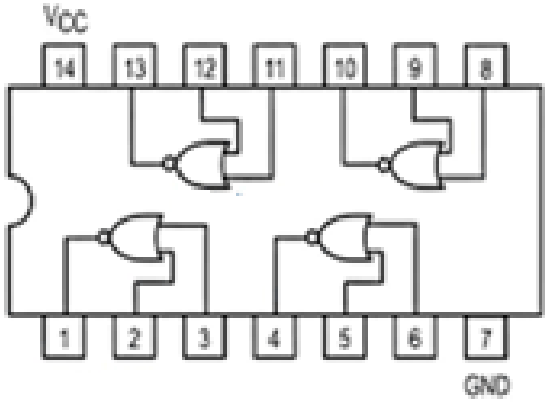


Fig. 3: Quad 2 Input NOR Gate (IC-7402)

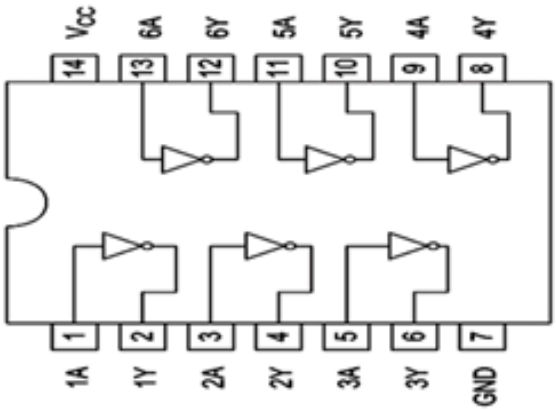


Fig 4: Hex Inverter (IC-7404)

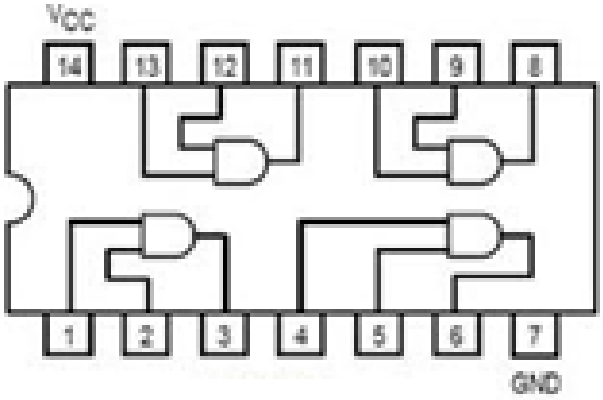


Fig. 5: Quad 2 Input AND Gate (IC-7408)

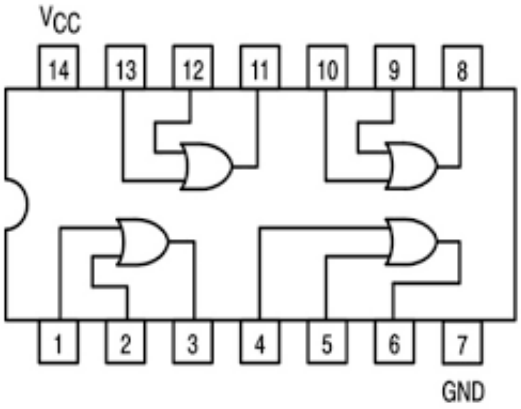


Fig 6: Quad 2 Input OR Gate (IC-7432)

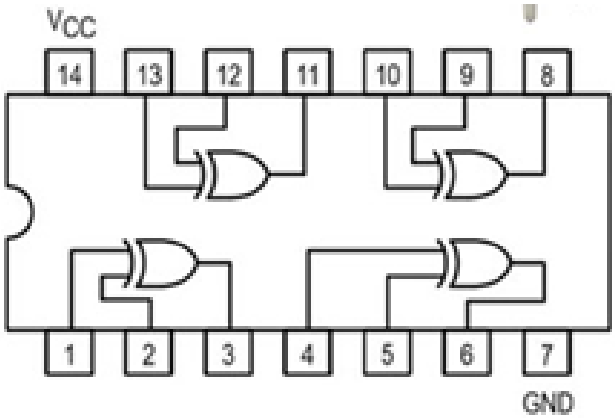
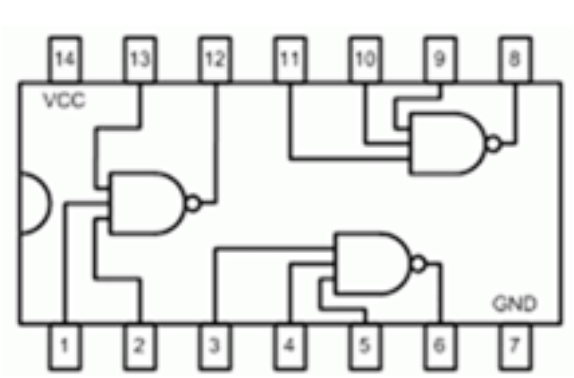


Fig. 7: Quad 2 Input XOR Gate (IC-7486)



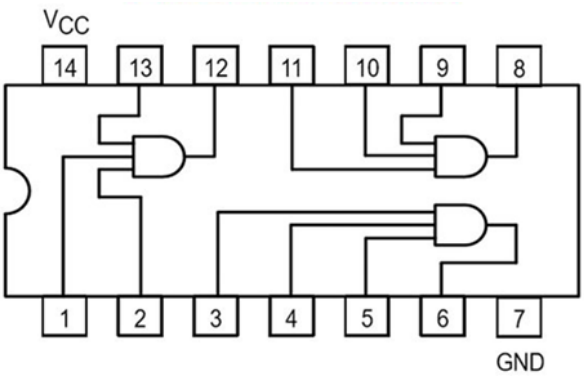


Fig 8: 3-Input NAND Gate (IC-7410)

Fig. 9: 3-Input AND Gate (IC-7411)

Example Implementation of a Logic Circuit:

Build a circuit to implement the Boolean function F = (A'.B')', please note that the notation A' refers to $\overline{\mathbf{A}}$. You should use that notation during the write-up of your laboratory experiments.

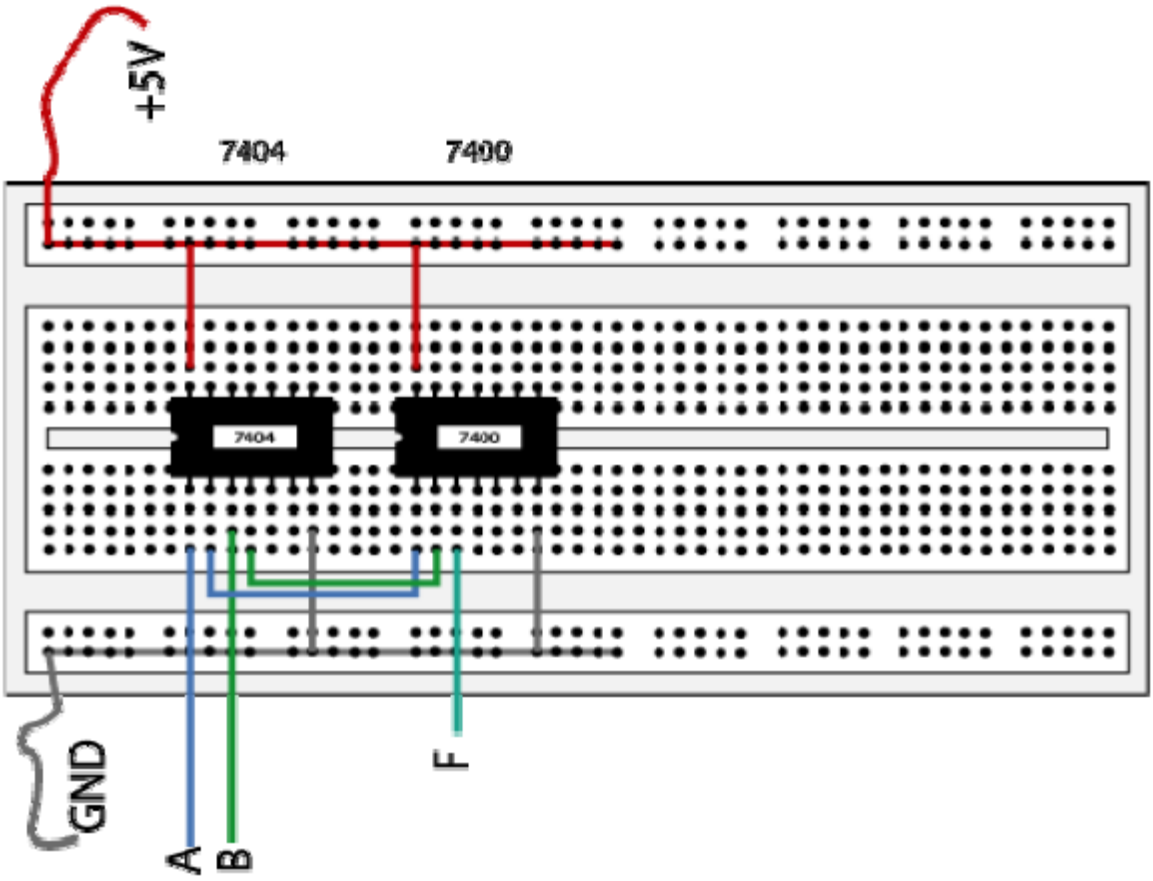


Fig 10. The complete designed and connected circuit

Sometimes the chip manufacturer may denote the first pin by a small indented circle above the first pin of the chip. Place your chips in the same direction, to save confusion at a later stage. Remember that you must connect power to the chips to get them to work.

Discussion of Gate parameters

Analog signal is defined as one whose amplitude is proportional to the quantity it represents. As this signal is continuous in time it has an infinite set of possible values.

A digital signal is one that represents between two discrete levels of voltage. These changes are abrupt. The most positive fixed voltage represents 1 state and the most negative fixed voltage represents 0 state.

Logic gates are the most basic type of digital circuit which consists of two or more inputs and one output. A gate can be used alone to perform a logic function. It can also be connected to several other gates to form a logic network.

Each type of basic logic gate is assigned with a symbol that has a distinct shape so that it can be distinguished from others and symbolically represent a logic gate.

A truth table is a formal specification that describes the exact circuit output behavior for every possible set of inputs. Truth tables are also called function tables.

In most situations logic signals applied to the inputs of logic gates are not stationary levels. Instead they are voltages that change continually between two states. To illustrate how logic gates respond to input logic signals in relation to time, timing diagrams are used. It is a plot of signal values/ amplitude Vs time.

An IC (integrated circuit) is a miniature chip of silicon or germanium on which an entire electronic circuit consisting of resistors, capacitors, diodes and transistors are built. When combined, these components are capable of performing many functions which include logic gate operations.

Several types of IC packages are available. A very popular package is DIP which stands for dual in – line package. “Dual in – line” means that the pins are positioned in a line on both sides of the IC package. The most common IC‟s have 8, 14, 16, 24 or 40 pins, which are numbered in a counter clockwise direction when viewed from the top. Pin number 1 is located next to an indentation or dot placed on the body of the IC package

To operate properly power must be supplied to the pin labeled VCC and ground must be connected to the pin labeled GND. Most of the pins are used as input and output connections for the circuitry.

The available digital IC‟s may be classified as – TTL (transistor – transistor logic) family, CMOS (complementary metal oxide semiconductor) family and ECL (emitter coupled logic) family. As most widely used gates have their equivalent CMOS gates sometimes it may suffice to conduct experiments only on TTL gate.

VIHmin(VILmin) is the minimum input voltage used to represent a logic high (logic low).
VIHmax (VILmax) is the maximum input voltage used to represent a logic high (logic low).
VOHmin(VOLmin) is the minimum output voltage used to represent a logic high (logic low).
VOHmax (VOLmax) is the maximum input voltage used to represent a logic high (logic low).

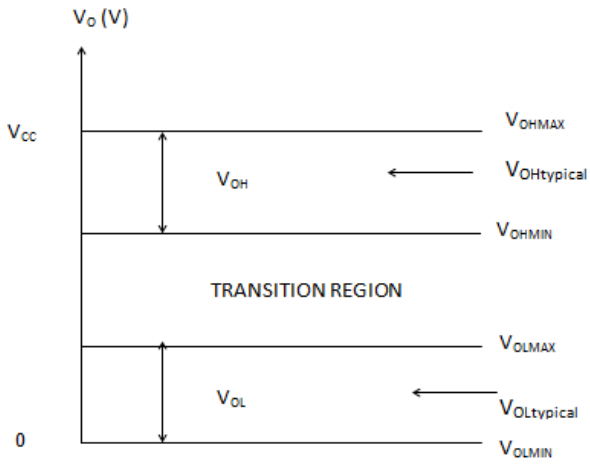


Fig 11: TTL output levels

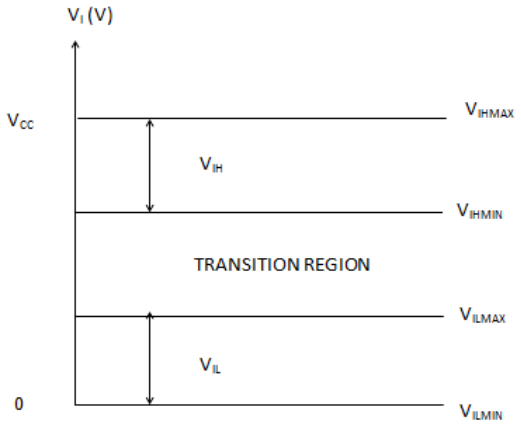


Fig 12: TTL input levels

Useful IC numbers and descriptions:

IC Number	Description	IC Number	Description
7400	Quad 2-NAND Gate	74112	Dual J-K F/F with preset and clear
7402	Quad 2-NOR Gate	74121	Mono-stable Multi-vibrator
7404	Hex Inverter	74123	Dual Mono-stable Multi-vibrator
7406	Hex Inverter, buffer	74125	Quad 3-state buffer
7408	Quad 2-AND gate	74138	Decoder
7410	3 input NAND gate	74139	Dual 1:4 De-MUX/Decoder
7411	Triple 3-AND gate	74147	10:4 Priority Encoder
7413	Dual 4-NAND Schmitt Trigger	74148	Priority encoder, Octal to Binary
7414	Hex Schmitt Trigger Inverter	74151	8:1 MUX
7420	4 input NAND gate	74153	Dual 4:1 MUX
7427	Triple 3-NOR gate	74154	1:16 DMUX
7432	Quad 2-OR Gate	74157	Quad 2:1 MUX
7447	BCD-Segment Decoder	74164	8 bit shift register
7473	Dual JK negative edge triggered flip-flop	74180	8 bit parity generator
7474	Dual D F/F	74192	Resettable Counter

7476	Dual J-K F/F	74193	4 bit binary counter
7483	4-bit Full adder	74194	Shift Register
7485	4-Bit Magnitude Comparator	74283	4 bit binary adder
7486	Quad 2-X-OR Gate	74365	Buffer/ Driver
7490	Decade Counter	DAC 0808	Digital to Analog Converter
7491	8-Bit Shift Register	ADC0804/09	Analog to Digital Converter
7492	Divide by 12 Counter	CD 4040	12 stage binary counter
7493	4-bit Binary Ripple Counter	CD4017	5 stage Jonson Counter
7495	4-Bit Shift Register	CD 4011	Quad 2 NAND gate
74180	Parity Generator	CD 4027	Dual JK flip-flop
7473	Dual J-K negative edge triggered Flip Flop	7410	Triple 3-input NAND Gate
74157	Quad 2:1 MUX	74138	Negative edge triggered dual J-K F/F

EXPERIMENT 1

Study of Implementation of Logic Gates

i) **Verification of truth table for basic logic gates (AND, OR, NOT, NAND, NOR, XOR)**

ii) **Study of Universality of NAND and NOR gates and realization of basic gates**

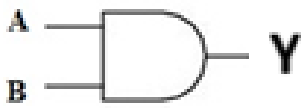
Aim:

- To get familiarization with IC trainer kit.
- To verify the characteristics of logic gates.
- To realize basic logic functions using universal gates.

APPARATUS REQUIRED: Power Supply, Digital Trainer Kit., Connecting Leads, IC's (7400, 7402, 7404, 7408, 7432, and 7486)

THEORY:

AND Gate: The AND operation is defined as the output as (1) one if and only if all the inputs are (1) one. 7408 is the two Inputs AND gate IC. A&B are the Input terminals & Y is the Output terminal.



Y = A.B

Truth Table		
A	**B**	**Y**
0	**0**	**0**
0	**1**	**0**
1	**0**	**0**
1	**1**	**1**

OR Gate: The OR operation is defined as the output as (1) one if one or more than 0 inputs are (1) one. 7432 is the two Input OR gate IC. A&B are the input terminals & Y is the Output terminal.

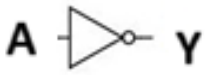
Y = A + B



Truth Table		
A	**B**	**Y**
0	**0**	**0**
0	**1**	**1**
1	**0**	**1**
1	**1**	**1**

NOT GATE: The NOT gate is also known as Inverter. It has one input (A) & one output (Y). IC No. is 7404. Its logical equation is, Y = A NOT B,

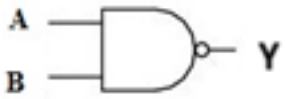
Y = A'



Truth Table	
A	**Y**
0	**0**
0	**1**

NAND GATE: The IC no. for NAND gate is 7400. The NOT-AND operation is known as NAND operation. If all inputs are 1 then output produced is 0. NAND gate is inverted AND gate.

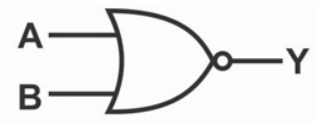
Y = (A. B)'



Truth Table		
A	**B**	**Y**
0	**0**	**1**
0	**1**	**1**
1	**0**	**1**
1	**1**	**0**

NOR GATE: The NOR gate has two or more input signals but only one output signal. IC 7402 is two I/P IC. The NOT- OR operation is known as NOR operation. If all the inputs are 0 then the O/P is 1. NOR gate is inverted OR gate.

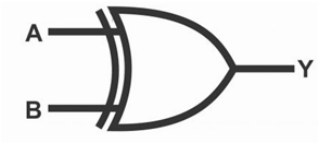
Y = (A+B)’



Truth Table		
A	**B**	**Y**
0	**0**	**1**
0	**1**	**0**
1	**0**	**0**
1	**1**	**0**

EX-OR GATE: The EX-OR gate can have two or more inputs but produce one output. 7486 is two inputs IC. EX-OR gate is not a basic operation & can be performed using basic gates.

Y = A B



Truth Table		
A	**B**	**Y**
0	**0**	**0**
0	**1**	**1**
1	**0**	**1**
1	**1**	**0**

ii) Study of Universality of NAND and NOR gates and realization of basic gates

- Basic Logic functions using NAND gate:

a) NOT Gate using NAND gate:

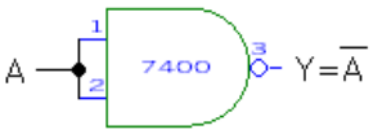


Fig. No. 1.1

b) AND gate using NAND gate:

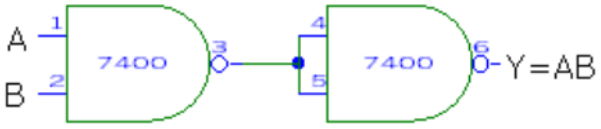


Fig. No. 1.2

c) OR gate using NAND gate:

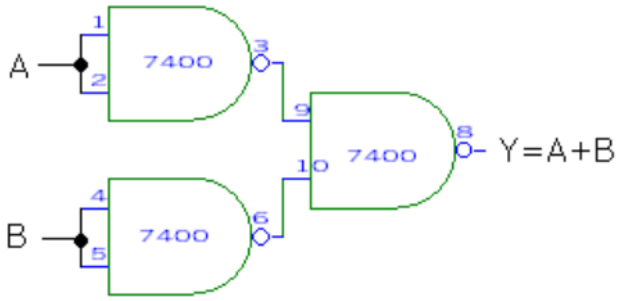


Fig. No. 1.3

- Basic Logic function using NOR gate:

a) NOT Gate using NOR gate:



Fig. No. 1.4

b) AND Gate using NOR gate

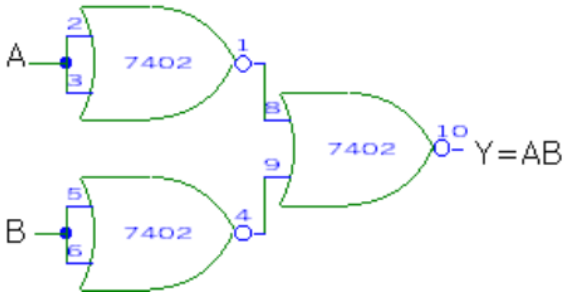


Fig. No. 1.5

c) OR gate using NOR gate

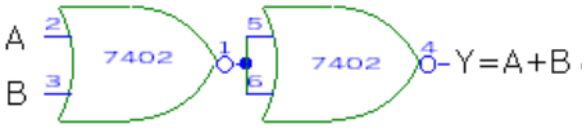


Fig. No. 1.6

EXPERIMENT 2

Study of basic arithmetic circuits.

Aim: To design, implement and test

i) Design and study of Half Adder/ Half Subtractor Half subtractor,
ii) Design and study of Full Adder/ Full Subtractor Full Subtractor.

Equipments & Components Required:

- **ICs 7408, 7432, 7404, 7400, 7486, 7410**
- **IC trainer kit, Connecting wires**

i) Half adder: A half adder circuit adds two bits to give the sum and carry outputs.

Inputs		Outputs	
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Table 2.1: Truth Table of Half Adder

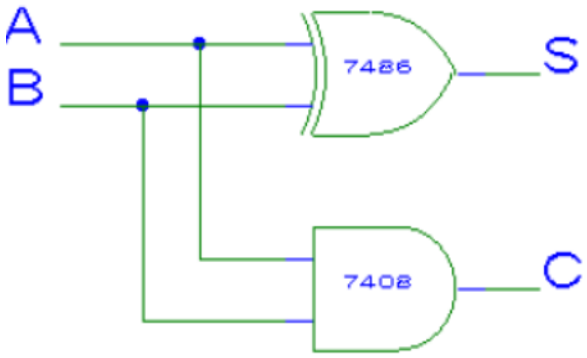


Fig. 2.1 Half Adder

$$S = \overline{A}.B + A.\overline{B}$$

$$C = A.B$$

ii) Half Subtractor

Inputs		Outputs	
X	Y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Table 2.2: Truth Table of Half Subtractor

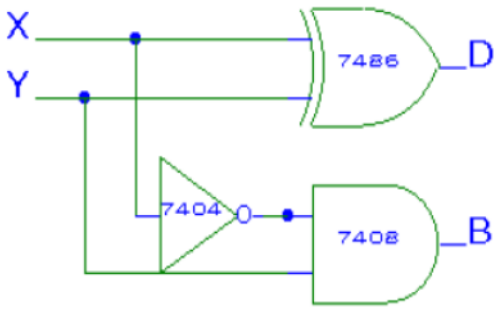


Fig. 2.2: Half Subtractor Circuit

D= A'B +AB'

B= A'B

iii) Full adder: Full adder adds two bits and also another bit called carry-in that is propagated from the previous stage. In effect, a full adder adds three bits to give out two outputs – sum and carry out. A, B and Cin are the three inputs, out of which Cin is to be considered as carry propagated from the previous stage. S and Cout are the three outputs.

Inputs			Outputs	
A	B	C_{in}	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$S = \overline{A}\overline{B}C_{in} + \overline{A}B\overline{C}_{in} + A\overline{B}\overline{C}_{in} + ABC_{in}$$
$$S = \overline{C}_{in}\left(\overline{A}B + A\overline{B}\right) + C_{in}\left(AB + \overline{A}\overline{B}\right)$$
$$S = \overline{C}_{in}\left(A \oplus B\right) + C_{in}\left(\overline{A \oplus B}\right)$$
$$S = \overline{C}_{in}\left(A \oplus B\right) + C_{in}\left(\overline{A \oplus B}\right)$$
$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + BC_{in} + AC_{in}$$

Table 2.3: Truth Table of Full Adder

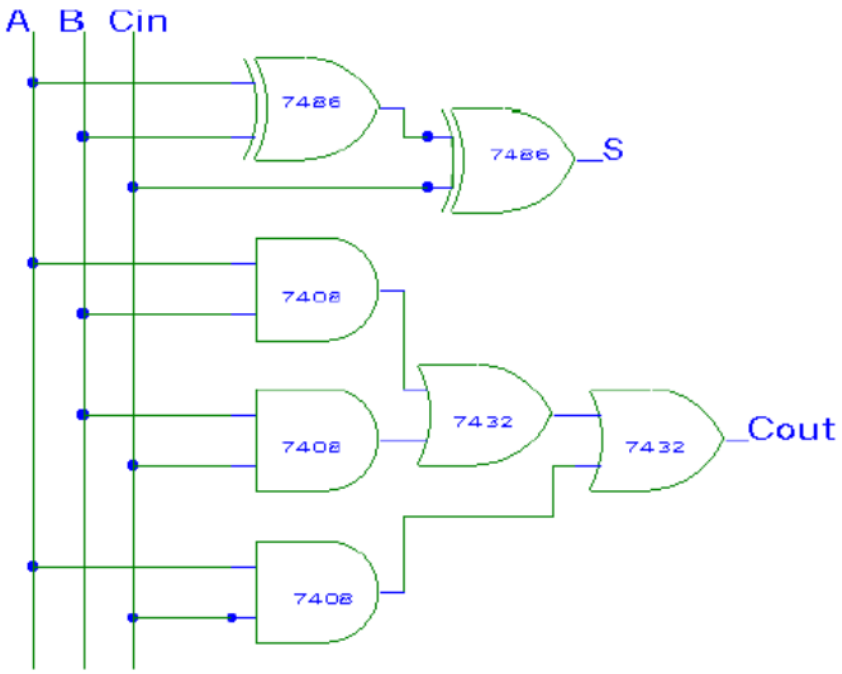


Fig. 2.3: Full Adder Circuit

iv) Full Subtractor

Inputs			Outputs	
A	B	b_i	D	b
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$D = b_i (AB + A'B') + b_i'(AB' + A'B)$$

$$= b_i (A \Theta B) + b_i' (A + B)$$

$$b = A'B + (A \Theta B) b_i$$

Table 2.4: Truth Table of Full Subtractor

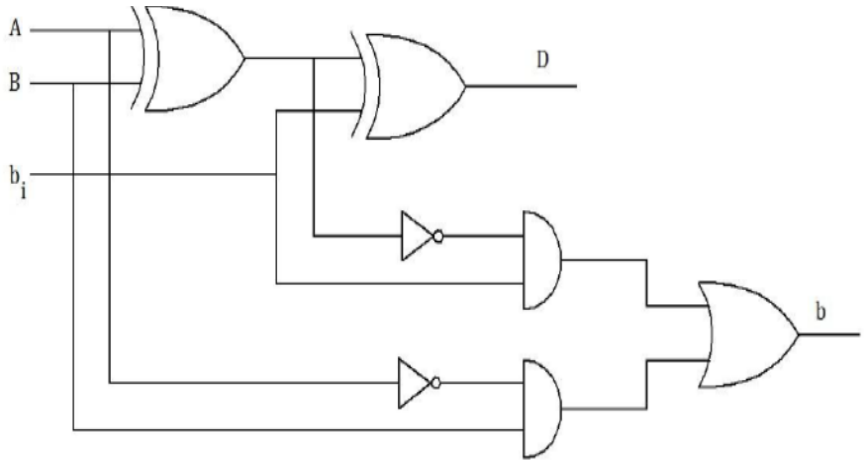


Fig. 2.4 : Full Subtractor Circuit

v) 4 –bit Binary full adder (IC 7483): The 7483 adds two 4-bit binary words plus the incoming carry .This contains 4 inter connected full adder and a look ahead carry adder circuit. Sum outputs are S3 to S0, the out-going carry is Cout.

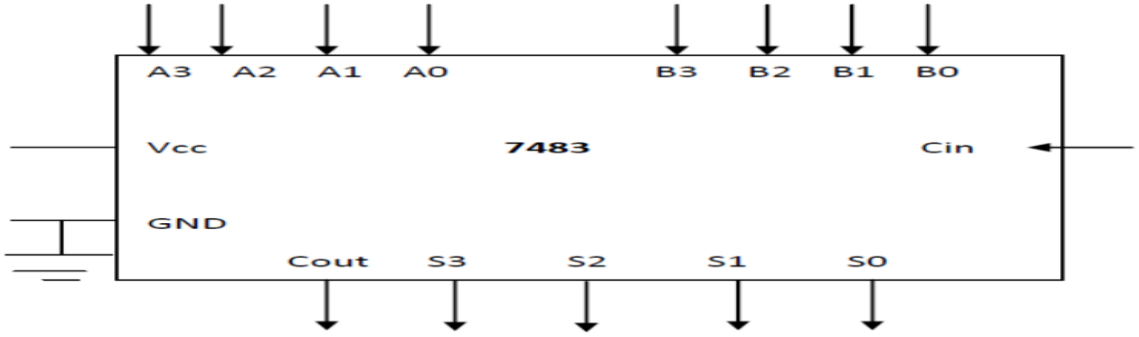


Fig. 2.5: Pin Diagram of IC 7485 (4-Bit binary full adder)

Procedure:

- Connect Vcc and Ground to the respective pins.
- Connect A3-A0 and B3–B0 to logic input switches.

- Connect Cout, S3-S0 to logic output LEDs.
- Give the inputs and note down the outputs.

- **4 Bit adder / 2's complement Subtractor**

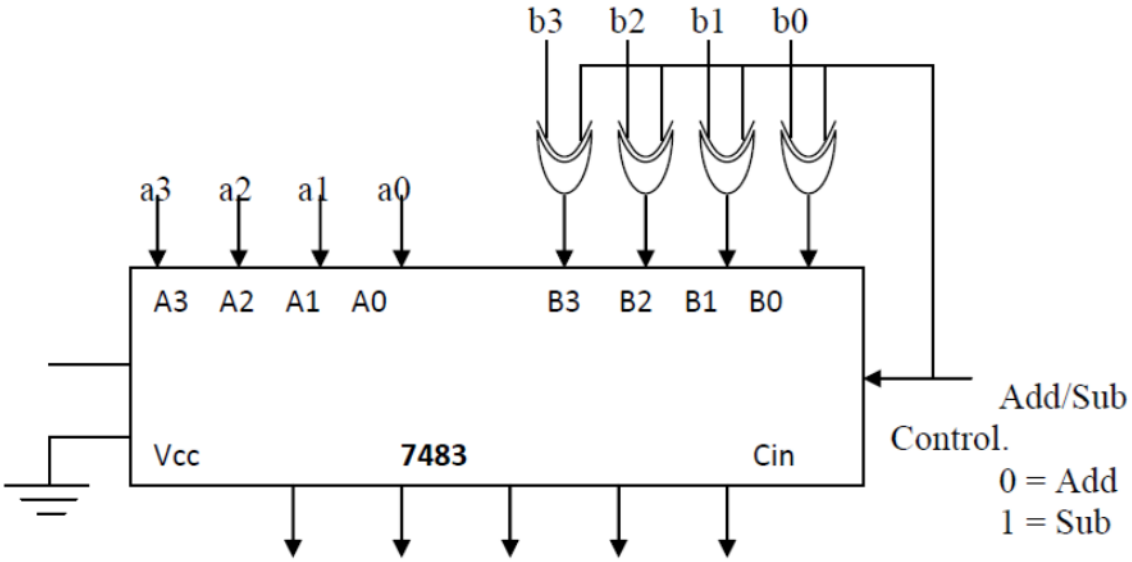


Fig. 2.6: Pin diagram / circuit for 4-bit 2'complement adder

Method:

- When control input is 0, input B = b3 b2 b1 b0 is added to input A = a3 a2 a1 a0.
- When control input is 1, 2's complement of B is added to input A. (i.e., A – B) Difference is available at S3 S2 S1 S0, borrow output at C_{out}.

- **BCD adder:**

When two 4 bit BCD numbers are added, if the sum exceeds 9 or if there is a carry ,then 6 is added to the sum and a carry is generated to the next decimal digit. Carry from the addition of 6 (if any) is neglected.

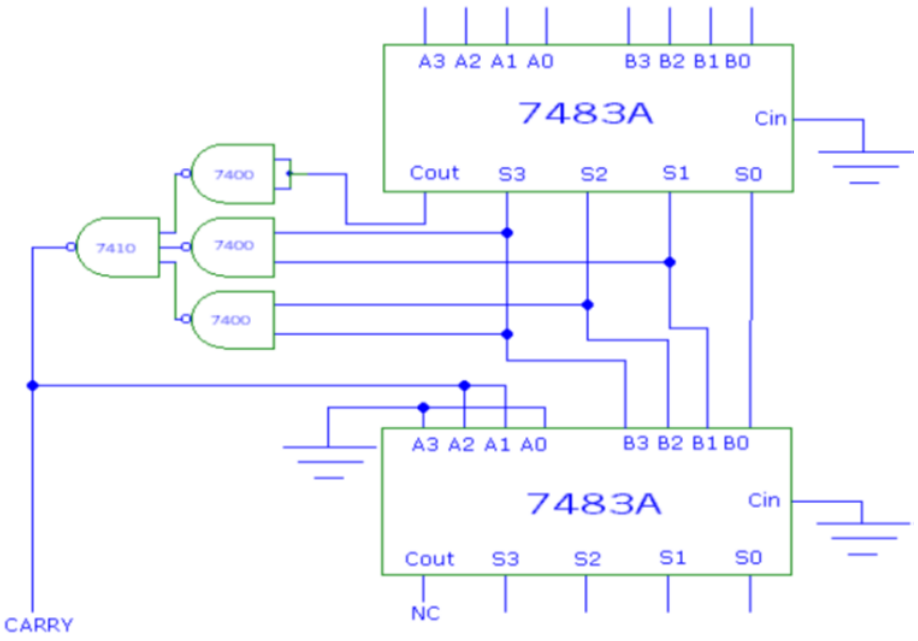


Fig. 2.7: BCD adder Circuit

Exercises:

1. Implement half adder circuit using only NOR gates.
2. Implement half Subtractor using only NAND gates.
3. Implement full adder using two half adder circuits and one OR gate.
4. Implement full Subtractor using two half Subtractors and any additional gate.
5. Implement BCD Subtractor using 7483 IC with result in 10‟s complement notation.
6. Design a 4bit XS-3 ADDER using 7483IC.

EXPERIMENT 3

DESIGN AND IMPLEMENTATION OF COMBINATIONAL CIRCUIT USING NAND AND NOR GATES

1. **To design, build and test simple combinational circuits using logic gates**

Aim: Realize the Boolean function f (W, X, Y, Z) = ∑m (1, 3, 5, 7, 8, 12, 14) using (i) basic gates (ii) only NAND gates (iii) only NOR gates.

Equipments & Components Required:

ICs required: IC7404, IC 7411, IC7408 and IC7432

Solution (using basic gates)

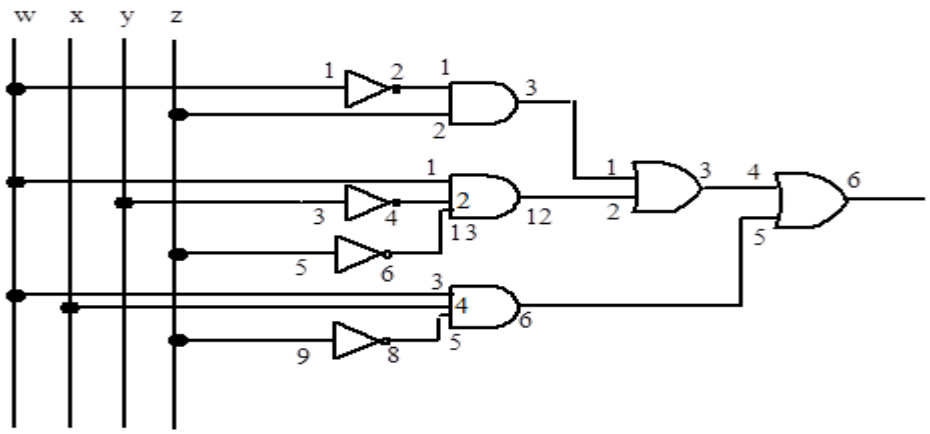


Fig. 3.1: solution for combinational circuit using basic gates.

Truth Table

Truth Table				
w	**x**	**Y**	**Z**	**Output**
0	**0**	**0**	**0**	
0	**0**	**0**	**1**	

EXPERIMENT 4

CODE CONVERTERS CIRCUITS

Aim:To design and implement the following code converters:

A. **4-bit Binary to Gray code**
B. **BCD to Excess-3 code**

Equipments & Components Required:

- ICs 7408, 7411, 7432, 7404, 7400, 7486, 7447, 7730
- IC trainer kit, Connecting wires

A. 4-bit Binary to Gray code:

Binary inputs	Gray outputs
$B_3B_2B_1B_0$	$G_3G_2G_1G_0$
0 0 0 0	0 0 0 0
0 0 0 1	0 0 0 1
0 0 1 0	0 0 1 1
0 0 1 1	0 0 1 0
0 1 0 0	0 1 1 0
0 1 0 1	0 1 1 1
0 1 1 0	0 1 0 1
0 1 1 1	0 1 0 0
1 0 0 0	1 1 0 0
1 0 0 1	1 1 0 1
1 0 1 0	1 1 1 1
1 0 1 1	1 1 1 0
1 1 0 0	1 0 1 0
1 1 0 1	1 0 1 1
1 1 1 0	1 0 0 1
1 1 1 1	1 0 0 0

Table 4.1: Truth table for 4-bit binary to grey code converter

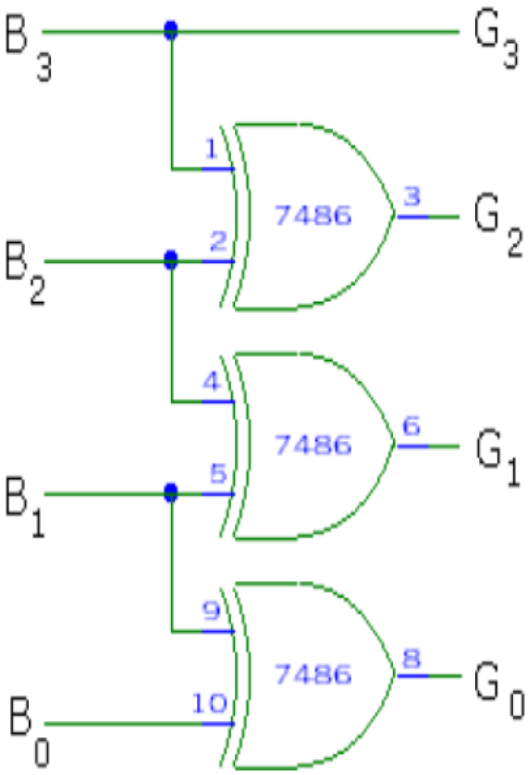


Fig. 4.1: 4-bit Binary to Gray code converter

$G_3 = B_3$
$G_2 = B_3 \oplus B_2$
$G_1 = B_2 \oplus B_1$
$G_0 = B_1 \oplus B_0$

Implement above circuit and verify the truth table

B. BCD to Excess-3 converter:

Excess-3 code is obtained by adding 3 to BCD number

First draw the truth table and then using K-map, get expression for each of Excess-3 bits.

BCD ABCD	Excess-3 WXYZ
0000	0011
0001	0100
0010	0101
0011	0110
0100	0111
0101	1000
0110	1001
0111	1010
1000	1011
1001	1100

$$W = A + BC + BD$$

$$X = \overline{B}D + B\overline{C}\,\overline{D} + A\overline{B}\overline{C}$$

$$Y = \overline{C}\,\overline{D} + CD$$

$$Z = \overline{D}$$

Table 4.2: Truth tale of BCD to grey code converter

Implement the above expressions and verify the truth table.

Exercises:

Design the following converters:

1. 4-bit Gray code to binary
2. 2421 BCD to 8421 BCD
3. BCD to 7-segment code

EXPERIMENT 5

MAGNITUDE COMPARATOR & PARITY CHECKER/GENERATER (ODD/EVEN PARITY)

Aim:
- To design, implement and study magnitude comparators.
- To study the working of 4 bit magnitude comparator IC 7485.

Equipments & Components Required:

- IC 7404, IC 7408, IC 7486, IC 7485
- IC trainer kit & Connecting wires

A. One bit magnitude comparator: A and B are two single bit inputs for the comparator and has three output lines, i.e. (A>B), (A<B), and (A=B).

- When input A is greater than input B, output (A>B) will be high
- When input A is less than input B, output (A<B) will be high
- When input A is equal to input B, output (A=B) will be high

Note that only one out of three outputs will be high at any time.

INPUTS		OUTPUTS		
A	B	A> B	A< B	A=B
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

Table 5.1: Truth table of Magnitude comparator

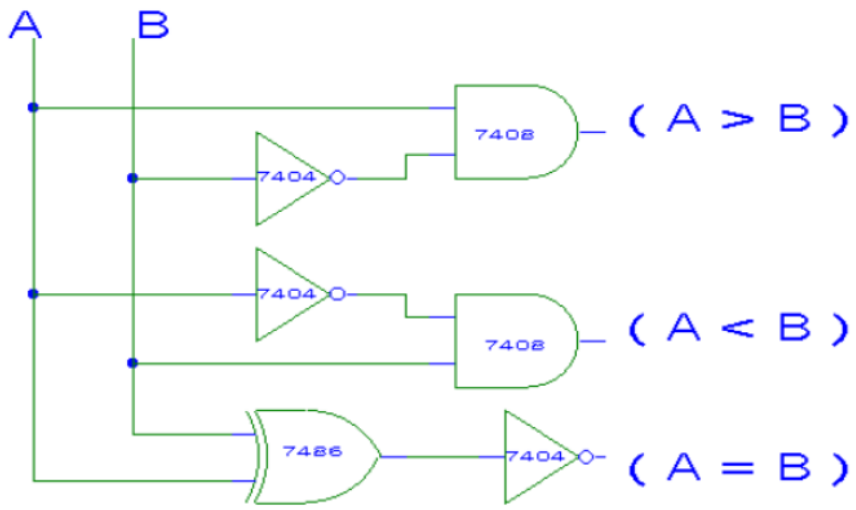


Fig. 5.1: Circuit for 1-bit Magnitude comparator

B. 4– bit magnitude comparator using IC 7485 : IC 7485 is a 4-bit magnitude comparator, which can compare two 4–bit binary or BCD codes. These IC‟s can be cascaded to compare words with more than 4 bits.

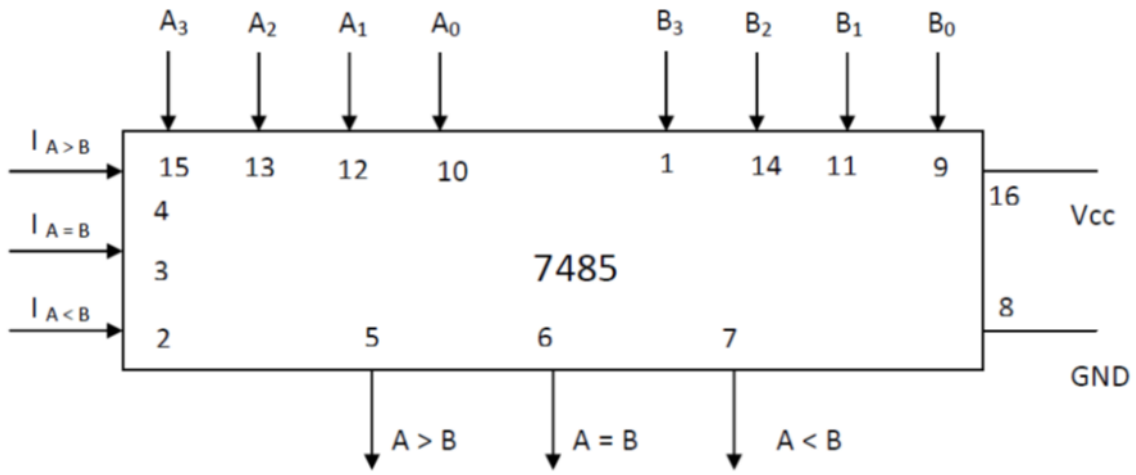


Fig. 5.3: Pin diagram of 4-bit Magnitude Comparator (IC 7485)

COMPARING INPUTS				CASCADE INPUTS			OUTPUTS		
A_3,B_3	A_2,B_2	A_1,B_1	A_0,B_0	$I_{A>B}$	$I_{A<B}$	$I_{A=B}$	A>B	A<B	A=B
$A_3>B_3$	X	X	X	X	X	X	1	0	0
$A_3<B_3$	X	X	X	X	X	X	0	1	0
$A_3=B_3$	$A_2>B_2$	X	X	X	X	X	1	0	0
$A_3=B_3$	$A_2<B_2$	X	X	X	X	X	0	1	0
$A_3=B_3$	$A_2=B_2$	$A_1>B_1$	X	X	X	X	1	0	0
$A_3=B_3$	$A_2=B_2$	$A_1<B_1$	X	X	X	X	0	1	0
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0>B_0$	X	X	X	1	0	0
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0<B_0$	X	X	X	0	1	0
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	1	0	0	1	0	0
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	0	1	0	0	1	0
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	0	0	1	0	0	1
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	X	X	1	0	0	1
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	1	1	0	0	0	0
$A_3=B_3$	$A_2=B_2$	$A_1=B_1$	$A_0=B_0$	0	0	0	1	1	0

Table 5.3: Truth table for 4-bit Magnitude Comparator

NOTE!!!!

• The upper part of the table describes normal condition operation that will occur in single device or in series expansion scheme.

• The lower part of the table describes feed forward conditions that exist in parallel expansion scheme.

D. Parity generator / checker

Aim:

- To design, implement and study 3-bit odd parity generator /checker
- To study the working of parity generator /checker IC 74180.

Equipments & Components Required:

- IC 7486, IC 74180
- IC trainer kit & Connecting wires

i) 3-bit odd parity generator/checker: An odd parity generator generates a 1 for an even number of 1's in the message, else 0.

Step 1: Consider a 3-bit message XYZ to be transmitted with an odd parity bit.

Step 2: Obtain an expression for P in terms of the message bits and realize using gates.

X	Y	Z	P Generated parity
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Table 5.4: Truth table for 3-bit parity generator (Odd parity)

$$P = \overline{X}\overline{Y}\overline{Z} + \overline{X}YZ + X\overline{Y}Z + XY\overline{Z}$$

$$P = \overline{X(Y \oplus Z)} + X(Y \oplus Z)$$

$$P = \overline{X \oplus (Y \oplus Z)}$$

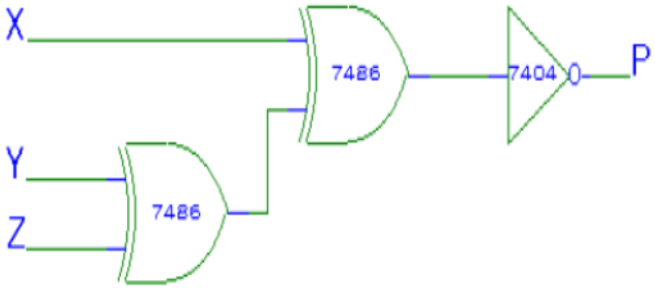


Fig. 5.4: Circuit for 3-bit parity generator (Odd parity)

Step 3: The odd parity checker consists of a 4-bit data consisting of X, Y, Z, P.

X	Y	Z	P	C (Output)
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Table 5.5: Truth table for 4-bit parity checker

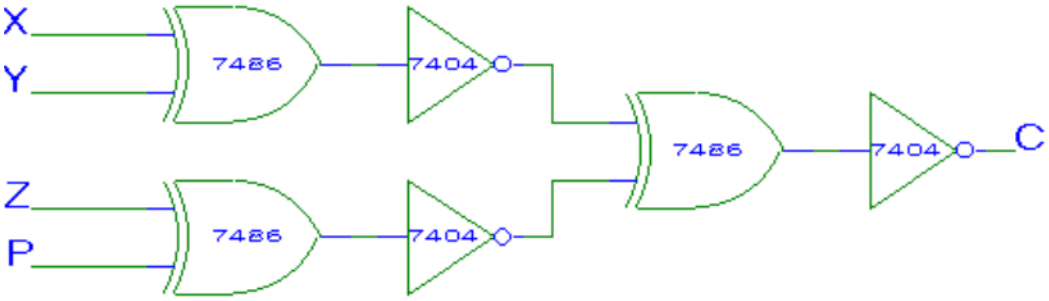


Fig. 5.5: Circuit for 4-bit parity checker

ii) Study of IC 74180 (parity generator/checker):

- The IC 74180 is a 9 bit (including the parity bit) party generator or checker, commonly used to detect errors in high speed data transmission or data retrieving systems.

- Both even and odd parity enable inputs and parity outputs are available, for generating or checking parity, on 8 bit of data.

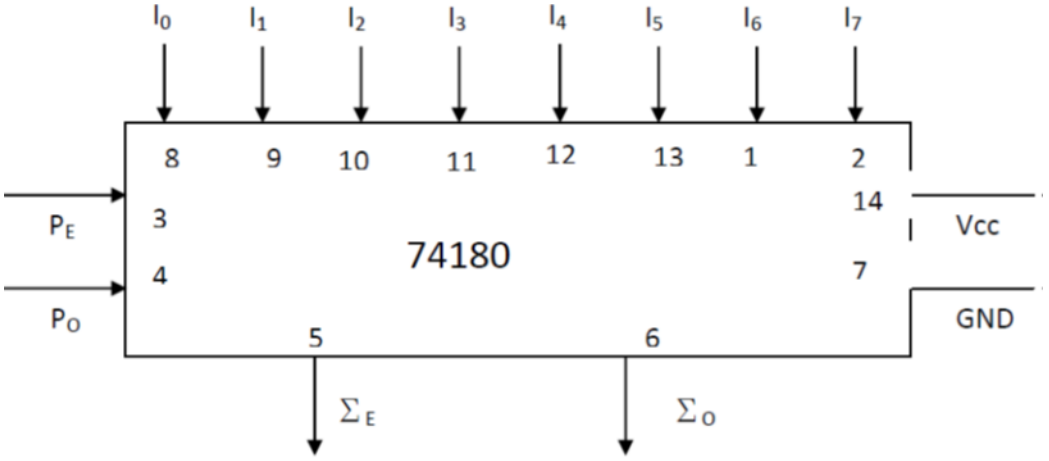


Fig. 5.6: Pin diagram of IC 74180

- The even parity output (ΣE) is high when an even number of data inputs (I0 – I7) are high, PE is high and PO is low.

- The odd parity output (ΣO) is high when an odd number of data inputs are high, PO is high and PE is low.

!!! NOTE: Observe and note down the outputs for the following inputs.

I_7 I_6 I_5 I_4 I_3 I_2 I_1 I_0	P_E P_O	$\sum_E$ $\sum_E$
1 1 1 1 0 0 0 0	1 0	
1 1 1 1 0 0 0 1	1 0	
1 0 1 0 1 1 0 0	0 1	
1 0 1 0 1 1 0 1	0 1	
0 0 0 0 0 0 0 0	1 0	
1 1 1 1 1 1 1 1	0 1	

Table 5.6: Observation Table

EXERCISES:

1. Design and implement 8-bit comparator using ICs 7485.

2. Design a 5-bit comparator using single IC 7485 and one gate.

3. Design and implement a 4 bit even parity generator/checker4. Design a 9 bit odd parity checker using a 74180 and an inverter.

EXPERIMENT 6

Multiplexers

MULTIPLEXERS – A multiplexer or data selector is a logic circuit that accepts several data inputs and allows only one of them at a time to get through the output. The routing of the desired data input to the output is controlled by SELECT inputs. The multiplexer acts like a digitally controlled multi position switch. The digital code applied to the SELECT inputs determines which data input to be switched to the output.

Aim:

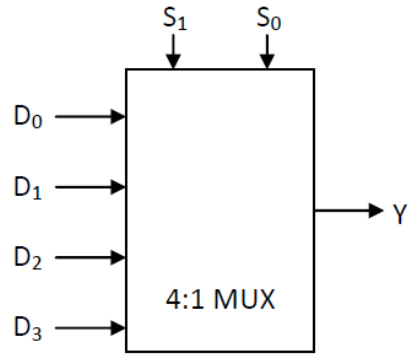
- To design and implement various multiplexer circuits and to generate logic functions using multiplexers.

Components and Equipment Required:

- ICs 74151, 74153
- Basic gates, universal gates
- IC trainer kit, connecting wires

i) **4:1 Multiplexer using only NAND gates**

Logic Symbol



$$\mathbf{Y} = \overline{\mathbf{S}}_1\overline{\mathbf{S}}_0\mathbf{D}_0 + \overline{\mathbf{S}}_1\mathbf{S}_0\mathbf{D}_1 + \mathbf{S}_1\overline{\mathbf{S}}_0\mathbf{D}_2 + \mathbf{S}_1\mathbf{S}_0\mathbf{D}_3$$

Fig. 6.1 - Logic Symbol

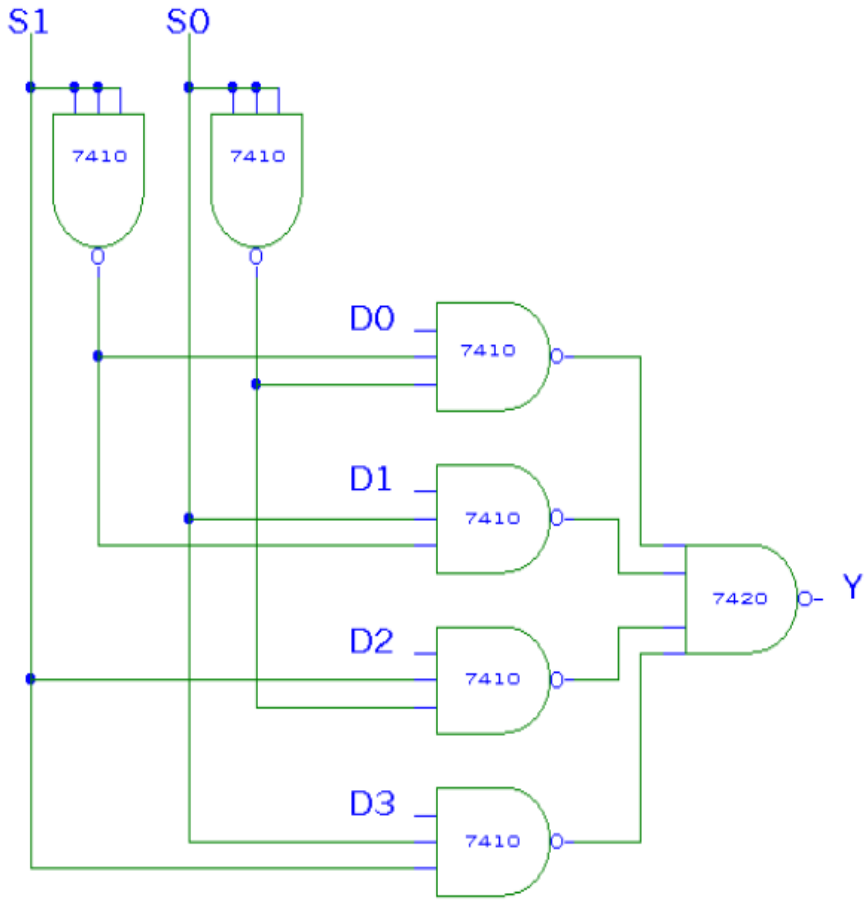


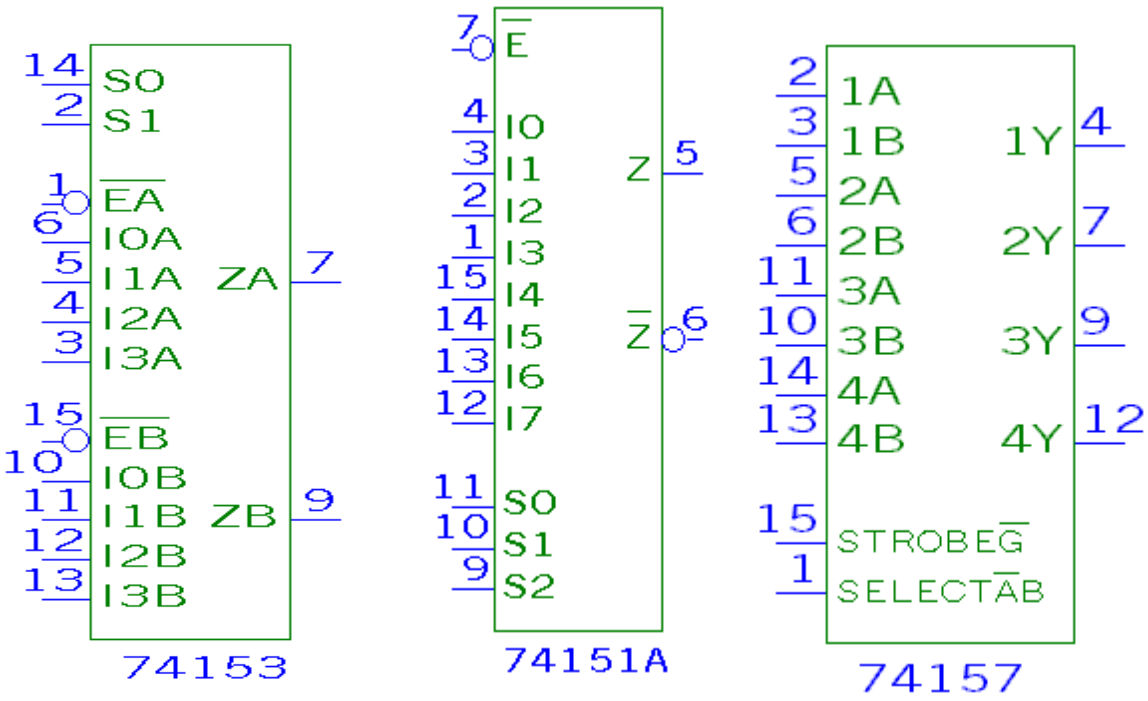
Fig. 6.2- Logic Diagram

Truth Table

S1	S0	Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Table 6.1- Truth Table of MUX

74153 – Dual 4:1 **Mux 74151 – 8:1** **Mux 74157 – Quad 2:1 Mux**



Note:

- Pin no.16 = Vcc, pin no.8 = GND for all the three ICs.
- For pin-details of the ICs, refer data manual.
- Test all the three ICs by giving appropriate inputs and observing the corresponding outputs. Thus verify the functional tables of these ICs.

ii) **Implement f = ∑m (0, 3, 5, 7, 8, 10, 14) using i) 8:1 MUX ii) 4:1 MUX and additional gates**

A	B	C	D	F	Using 8:1 MUX	Using 4:1 MUX
0	0	0	0	1	$I_0 = \overline{D}$	$I_0 = \overline{C \oplus D}$
0	0	0	1	0		
0	0	1	0	0	$I_1 = D$	
0	0	1	1	1		
0	1	0	0	0	$I_2 = D$	$I_1 = D$
0	1	0	1	1		
0	1	1	0	0	$I_3 = D$	
0	1	1	1	1		
1	0	0	0	1	$I_4 = \overline{D}$	$I_2 = \overline{D}$
1	0	0	1	0		
1	0	1	0	1	$I_5 = \overline{D}$	
1	0	1	1	0		
1	1	0	0	0	$I_6 = 0$	$I_3 = C\overline{D}$
1	1	0	1	0		
1	1	1	0	1	$I_7 = \overline{D}$	
1	1	1	1	0		

Table 6.2

using 8:1 MUX

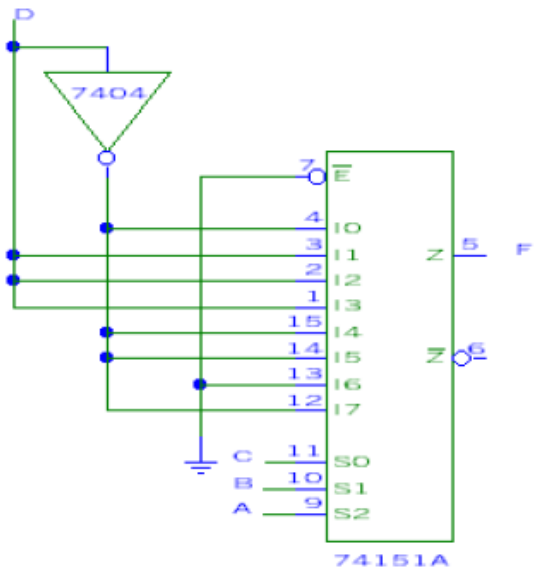


Fig. 6.3

using 4:1 MUX

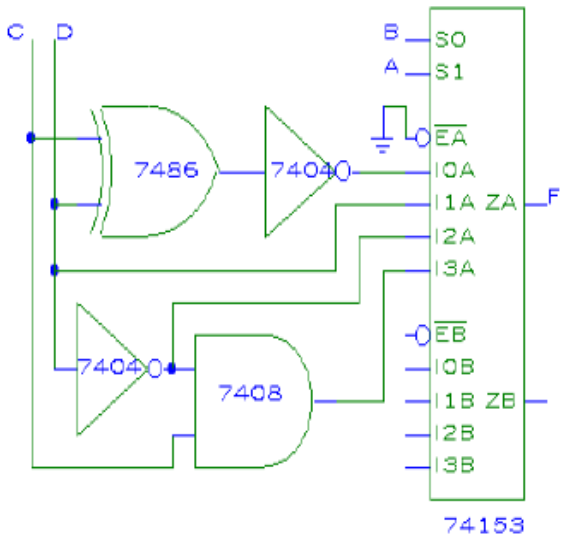


Fig. 6.4

Exercises:

1. Implement full adder using 4:1 MUX and 8:1 MUX
2. Implement F = ∑ m (0,5,7,11,15,16,18,25,29) using two 8:1 and one 2:1 MUX

EXPERIMENT 7

De-multiplexers

DEMULTIPLEXER – A de-multiplexer or data distributor is a circuit that receives information on a single line and transmits this information on one of the 2n possible output lines. The selection of a specific output line is controlled by n selection lines. A decoder can function as de-multiplexer if enable line is taken as data input.

Aim:

- To design and implement various multiplexer circuits using logic gates.
- To design and implement 1:8 De-multiplexer using IC

Components and Equipment Required:

- ICs 74151, 74153
- Basic gates, universal gates
- IC trainer kit, connecting wires

i) Design a 1 to 4 De-multiplexer using basic gates

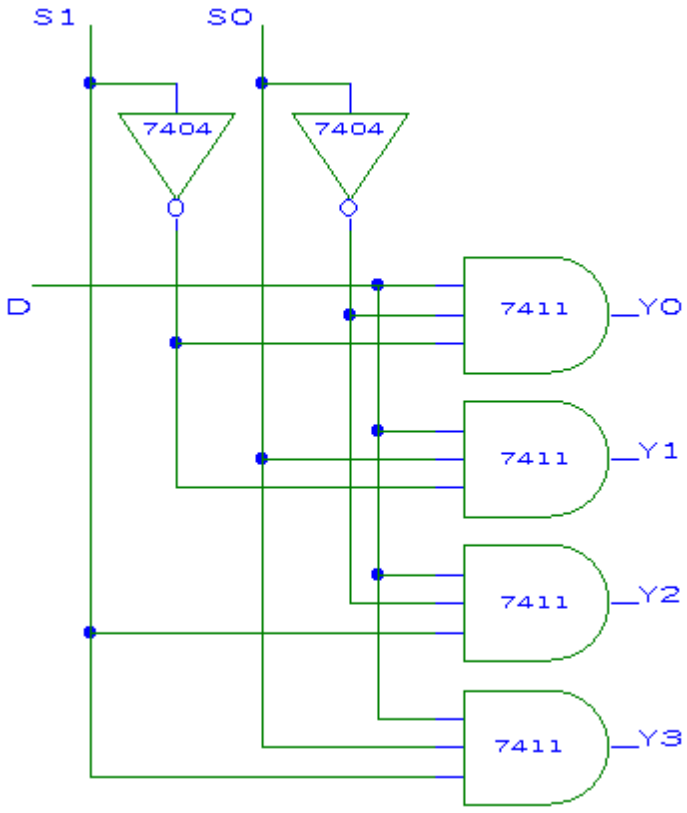


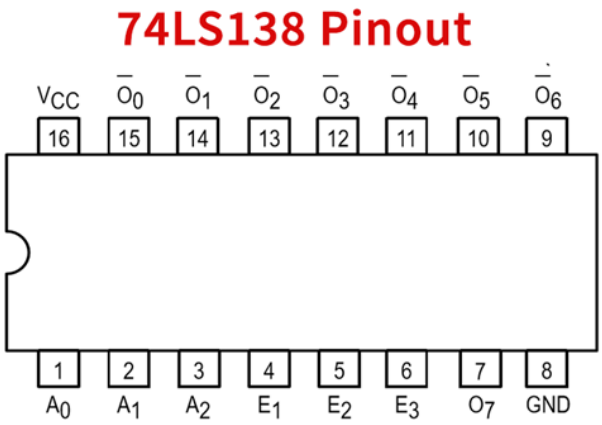
Fig. 7.1- Logic Diagram

Select lines		Data	Outputs			
S_1	S_0	D	Y_0	Y_1	Y_2	Y_3
0	0	0 / 1	D	0	0	0
0	1	0 / 1	0	D	0	0
1	0	0 / 1	0	0	D	0
1	1	0 / 1	0	0	0	D

Table 7.1

ii) To design and implement 1:8 De-multiplexer using IC

1:8 Demultiplexer multiplexer consists of 8 outputs, one inputs and three control or select lines . One data input line E of this multiplexer has been connected simultaneously with all the gates. These three gates remain enable or work alternately according to the address provided by all 3 select lines A0, A1, A2.



74LS138 Pin Configuration

Pin No	Pin Name	Description	Pin No	Pin Name	Description
1	A0	Address Input Pin A0	9	O6′	Active Low Output pin O6
2	A1	Address Input Pin A1	10	O5′	Active Low Output pin O5
3	A2	Address Input Pin A2	11	O4′	Active Low Output pin O4
4	E1′	Active Low Enable Input E1′	12	O3′	Active Low Output pin O3
5	E2′	Active Low Enable Input E2′	13	O2′	Active Low Output pin O2
6	E3	Active high Enable Input E3	14	O1′	Active Low Output pin O1
7	O7′	Active Low Output pin O7	15	O0′	Active Low Output pin O0
8	GND	Ground Pin	16	Vcc	Supply Voltage

EXPERIMENT 8

ENCODERS AND DECODERS

ENCODER – An encoder has 2^n (or less) input lines and **n** output lines. The output lines generate the binary code for 2^n inputs.

DECODER – It is a logic circuit that converts an N-bit binary input code into M output lines such that only one line is activated for each one of those possible combinations of inputs.

Aim:

- To design and implement Decoders and Encoders.

Components and Equipments Required:

- ICs 74139, 74138, 74154, 74148
- Basic gates, universal gates
- IC trainer kit, connecting wires

i) Design and implement 2 to 4 decoder using basic gates:

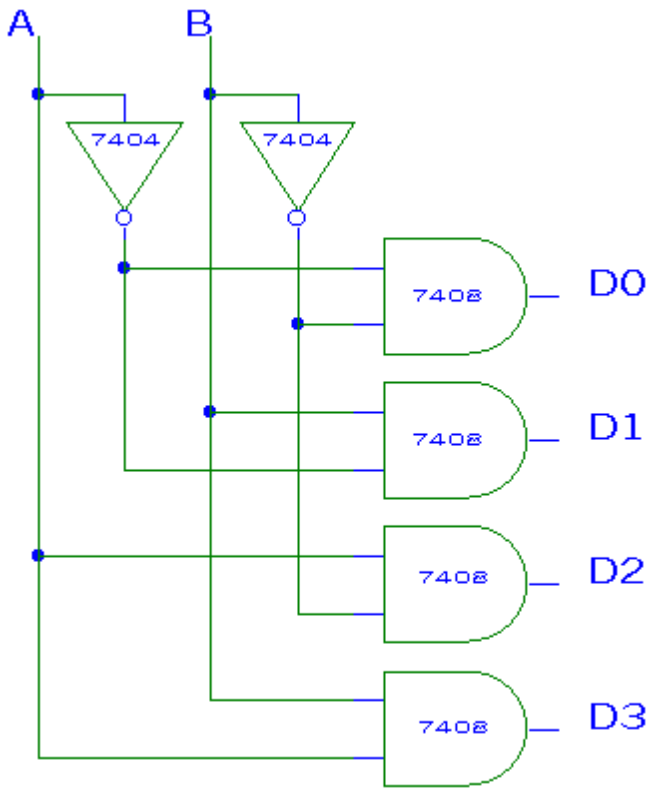


Fig. 8.1- Logic Diagram

INPUTS		OUTPUTS			
A	B	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Table 8.1

ii) Design and implement 2 to 4 decoder with enable input using only NAND gates

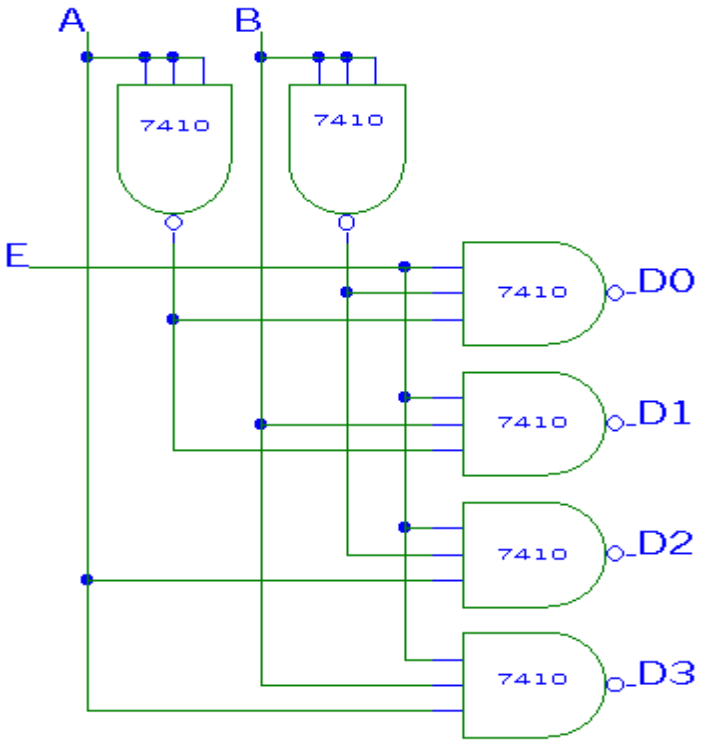


Fig. 8.2- 2 to 4 decoder using NAND Gate

Inputs			Outputs			
E	A	B	D_0	D_1	D_2	D_3
0	X	X	1	1	1	1
1	0	0	0	1	1	1
1	0	1	1	0	1	1
1	1	0	1	1	0	1
1	1	1	1	1	1	0

Table 8.2

EXPERIMENT 09

STUDY OF FLIP-FLOPS

Aim:

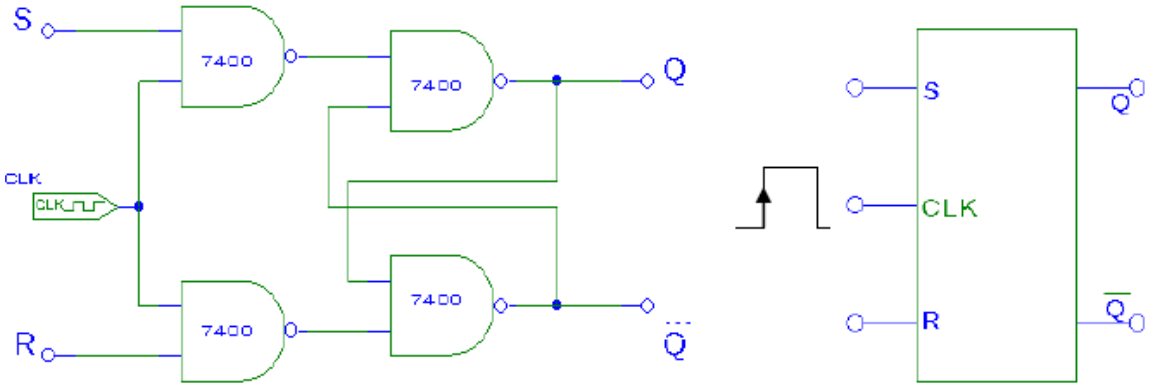
- To study the operations of SR Flip-flop, D Flip-flop, JK Flip-flop, T Flip-flop.
- To study the IC –7474 & IC – 7473.

Components and Equipment Required:

- ICs 7400, 7473, 7474
- IC trainer kit, connecting wires
- Cathode Ray Oscilloscope

Note: Refer the IC data manual, draw the block & pin diagram of ICs 7473 & 7474.

i) **SR Flip-flop using NAND gates:**

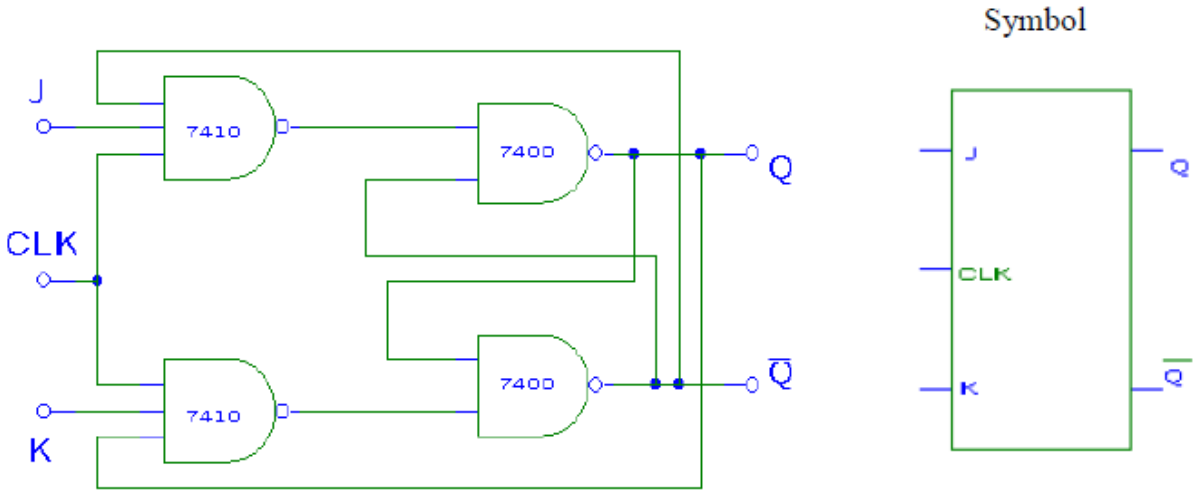


Note : Apply high-going mono pulse to CLK

INPUTS			OUTPUTS		OBSERVATIONS	
CLK	S	R	Q_{n+1}	$\overline{Q_{n+1}}$	Q_{n+1}	$\overline{Q_{n+1}}$
⊓	0	0	Q_n	$\overline{Q_n}$		
⊓	0	1	0	1		
⊓	1	0	1	0		
⊓	1	1	Not Allowed			

Fig. 9.1

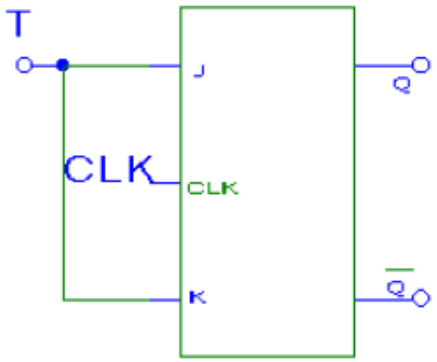
ii) **JK Flip-Flop using NAND gates:**



INPUTS			OUTPUTS	
CLK	J	K	Q_{n+1}	$\overline{Q_{n+1}}$
↑	0	0	Q_n	$\overline{Q_n}$
↑	0	1	0	1
↑	1	0	1	0
↑	1	1	$\overline{Q_n}$	Q_n

Fig. 9.2

iii) **T-Flip Flop using NAND gates:**



INPUTS		OUTPUTS	
CLK	T	Q_{n+1}	$\overline{Q_{n+1}}$
LOW	X	Q_n	$\overline{Q_n}$
↑	0	Q_n	$\overline{Q_n}$
↑	1	Toggling	

Fig. 9.3

EXPERIMENT 10

Aim:

- Study of IC 7474: Positive edge triggered dual D flip-flops with preset and clear.
- Study of 74LS73A:Negative edge triggered dual JK Flip-flops with clear
- Conversion of Flip Flops

Components and Equipment Required:

- ICs 7400, 7473, 7474
- IC trainer kit, connecting wires
- Cathode Ray Oscilloscope

Note: Refer the IC data manual, draw the block & pin diagram of ICs 7473 & 7474.

i) Study of IC 7474: Positive edge triggered dual D flip-flops with preset and clear.

NOTE: Draw the pin diagram of 7474 IC from the manual

INPUTS				OUTPUTS	
PRESET	CLEAR	CLOCK	DATA	Q_{n+1}	$\overline{Q_{n+1}}$
PR	CLR	CLK	D		
0	1	X	X	1	0
1	0	X	X	0	1
1	1	0	X	Q_n	$\overline{Q_n}$
1	1	↑	1	1	0
1	1	↑	0	0	1

Table No. 10.1

ii) Study of 74LS73A: Negative edge triggered dual JK Flip-flops with clear

NOTE: Draw the pin diagram of IC 7473 from the manual

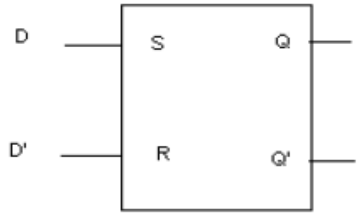
INPUTS				OUTPUTS	
CLEAR $\overline{CLR}$	CLOCK CLK	J	K	Q_{n+1}	$\overline{Q_{n+1}}$
0	X	X	X		
1	↓	0	0		
1	↓	1	0		
1	↓	0	1		
1	↓	1	1		
1	1	X	X		

Table No. 10.2

iii)**Conversion of flip flops:**

(a) Convert SR flip flop to D flip flop

D	**Q**	**Q+**	**S**	**R**
0	0	0	0	D
0	1	0	0	1
1	0	1	1	0
1	1	1	d	0



Using K Map we get the expression for

S= D

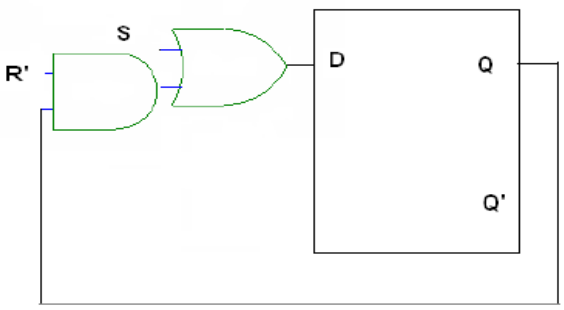
R=D'

Note: Build SR flipflop using Nand gates

Fig. 10.1

(b) Convert D to SR Flip flop

S	R	Q	Q+	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	X	X
1	1	1	X	X



Using K map we get the expression for D

D=R'Q + S

Fig. 10.2

Exercise:

1. Convert T flip flop to JK flip flop
2. Convert D flip flop to T flip flop
3. Convert JK flip flop to SR flip flop.
4. Construct SR flip flop using NOR gates

EXPERIMENT 11

COUNTERS

Aim:

- To design and test 4-bit Synchronous up counter using IC7473.
- To test Johnson and Ring counters.

Components and Equipment Required:

- ICs 7408, 7473, 7474
- IC trainer kit, connecting wires
- Cathode Ray Oscilloscope

Note: Refer the IC data manual, draw the block & pin diagram of ICs 7473 & 7474.

i) 4-bit Synchronous up – counter:

Excitation Table of J K Flip-Flop:

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

- In the Excitation table shown above Q_n and Q_{n+1} are present and next state outputs. For a 4 bit counter, it is required to count from 0 to 15, which requires 4 J K Flip-Flops.
- Let the outputs of these Flip-Flops be Q_D,Q_C,Q_B,Q_A respectively and J_A,K_A be the inputs of Flip-Flop A. J_B,K_B of Flip-Flop B & so on.

State Table:

Q_D	Q_C	Q_B	Q_A	Q_{D+1}	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_D	K_D	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	X	0	0	X	0	X	1	X
1	0	0	1	1	0	1	0	X	0	0	X	1	X	X	1
1	0	1	0	1	0	1	1	X	0	0	X	X	0	1	X
1	0	1	1	1	1	0	0	X	0	1	X	X	1	X	1
1	1	0	0	1	1	0	1	X	0	X	0	0	X	1	X
1	1	0	1	1	1	1	0	X	0	X	0	1	X	X	1
1	1	1	0	1	1	1	1	X	0	X	0	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	1	X	1	X	1

Draw K-maps for various Js & Ks and simplify. Implement the logic circuit and verify.

NOTE: To implement the circuit, two nos. of 7473 and one no. of 7408 ICs are required.

For IC7473 : Vcc is pin 4, Gnd is pin 11
For IC7408 : Vcc is pin 14 , Gnd is pin 7

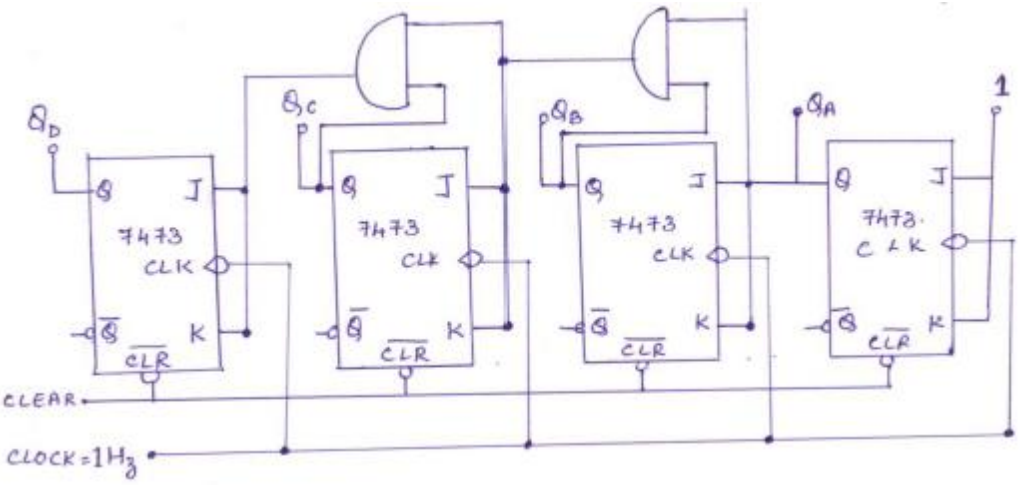


Fig. 11.1

Procedure:

1. Connect the circuit.
2. Momentarily clear all the inputs by giving Cr = 0 & then make it permanently high by giving Cr = 1.
3. Verify the circuit by observing the output for each clock pulse. After the 15th clock pulse (i.e. for the 16th clock pulse) output DCBA = 0000

Tabular Column:

CLOCK PULSE	D	C	B	A
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Table . 11.1

Johnson and Ring counters:

PROCEDURE **:**

1. Connect the circuit as shown in the circuit diagram. Connect the clear input to logic level switch and Q_3-Q_0 to LED indicators.
2. Clear all the Flip Flops momentarily by giving clear Cr=0 & then make it„1" permanently. Observe the output for each clock pulse.

Ring Counter:

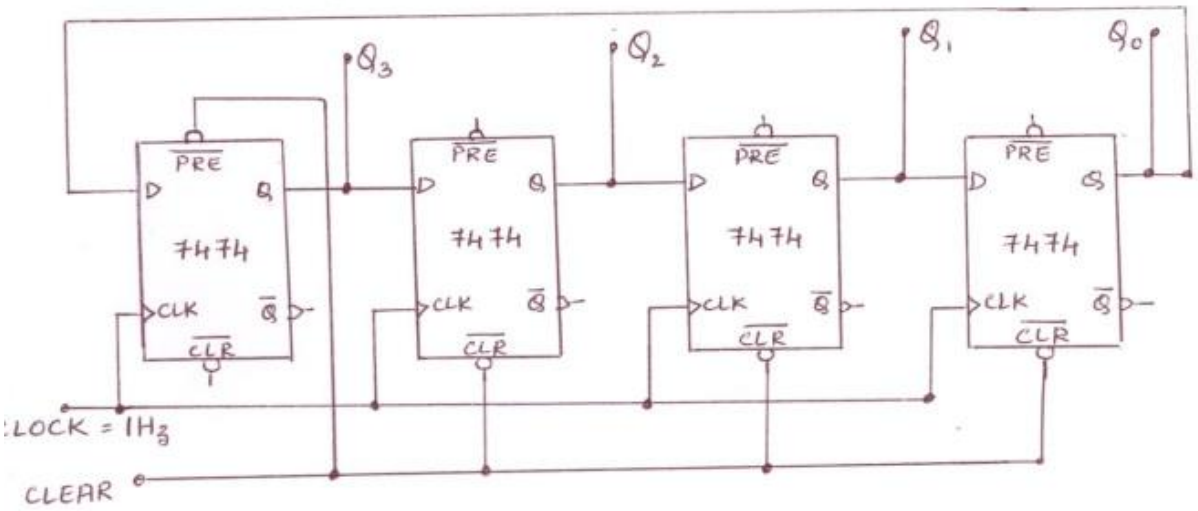


Fig. 11.2

Johnson Counter:

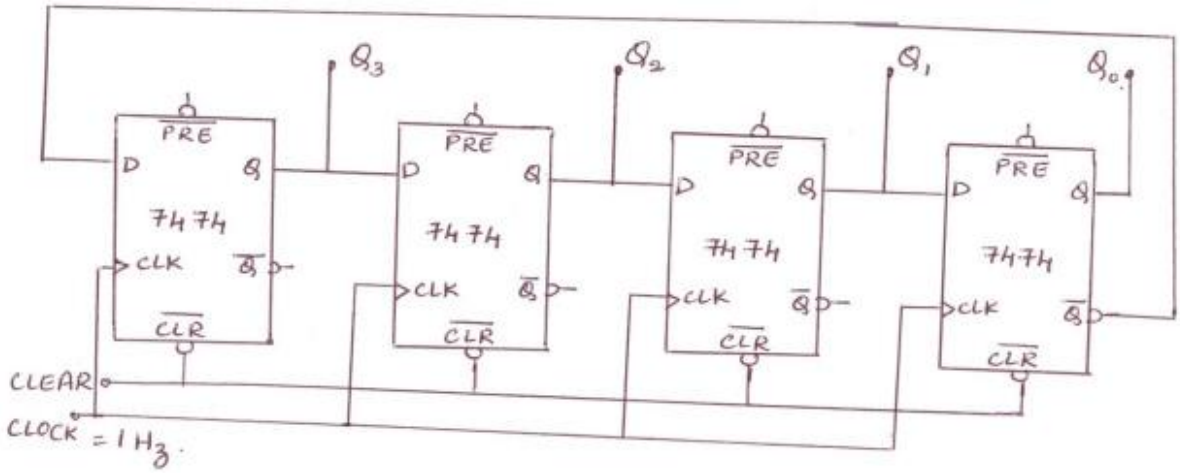


Fig. 11.3

Tabular Columns:

Johnson's counter

CLOCK PULSE	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

Table: 11.2

Ring counter

CLOCK PULSE	Q_3	Q_2	Q_1	Q_0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0

Table : 11.3

Exercises:

1. Design and implement a decade counter and verify its operation
2. Design of synchronous up-down counter

EXPERIMENT 12

SHIFT REGISTERS

AIM: To design and test the following Shift register topologies.

- Serial-in Serial-out
- Serial-in Parallel-out.

COMPONENTS REQUIRED:

IC 7474 (2Nos), IC 7408, IC 7400, IC 7495, IC 74164
IC Trainer kit, connecting wires

Circuit diagram:

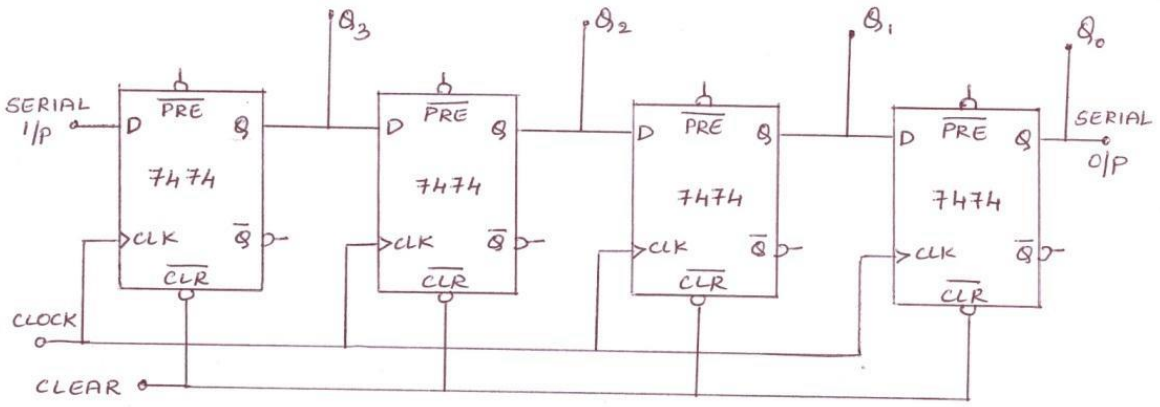


Fig. 12.2

PROCEDURE:

A. FOR SISO, SIPO SHIFT REGISTER:

1. Connect the circuit as shown in the circuit diagram
2. Switch on & momentarily clear all the Flip-Flops by giving low to the clear input. Then make it permanently logic1.
3. Enter any 4-bit serial input using clock pulse for every bit.
4. for example, to enter 1011
5. First make serial input =1, apply a clock pulse,
6. Make serial input=1, again apply a clock pulse,
7. Make serial input=0, again apply a clock pulse,
8. Make serial input =1, apply a clock pulse.
9. This ensures that the register has word 1011 stored in it.
10. Observe the data at the output of each shift register simultaneously.
11. This constitutes parallel output data.
12. To observe the output data in serial fashion, apply clock pulse by keep in serial input data zero.

AIM: To design and test the following Shift register topologies.

- Parallel-in Serial-out.

- Parallel-in Parallel-out.

COMPONENTS REQUIRED:

- IC 7474 (2Nos), IC 7408, IC 7400, IC 7495, IC 74164
- IC Trainer kit, connecting wires

Circuit diagram:

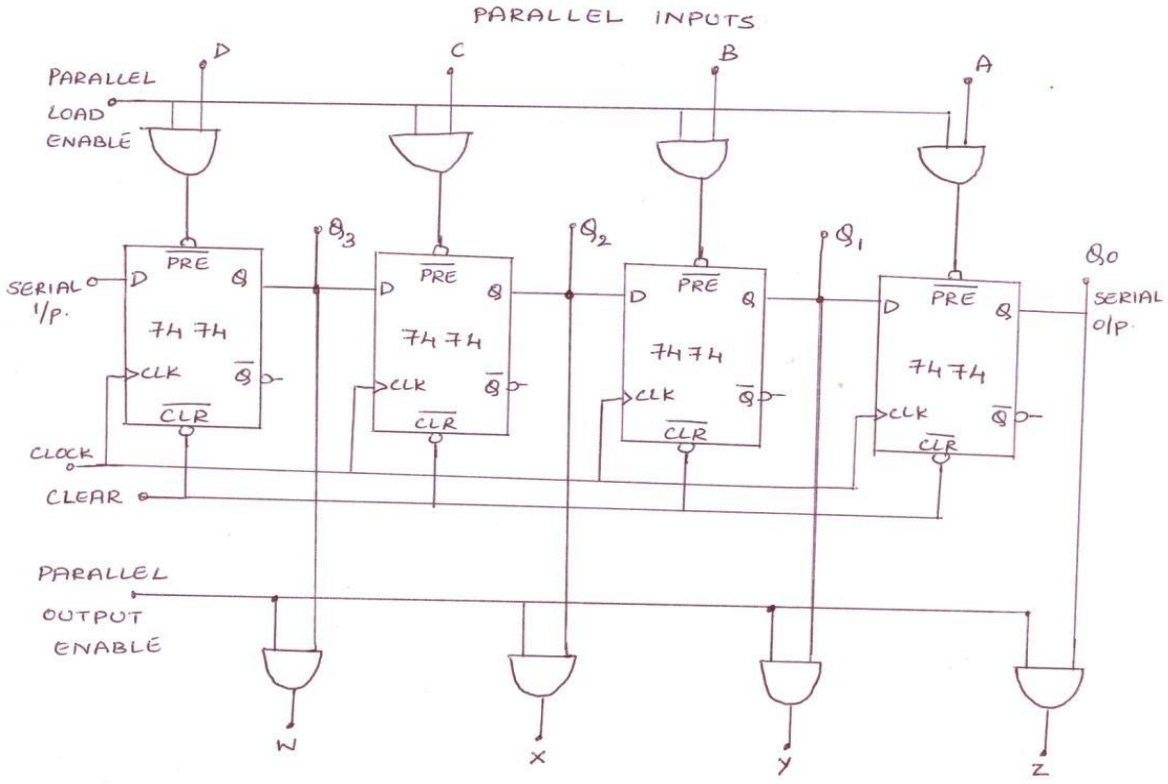


Fig. 12.3

FOR PISO, PIPO SHIFT REGISTER:

1. Make the clear and parallel enable low i.e. *CLR=0, PE=0.*

2. To load the input data in parallel fashion, apply 4 bit data(*D C B A*) and make the parallel enable *PE=1*.

3. To read the output data in parallel fashion, make parallel output enable =1, i.e. *PE=1*.

4. To read the output data in serial fashion, make parallel output enable =0 i.e. *PE=0* &*CLR=1*. Then apply clock pulse making serial input data=0. Data gets shifted from left Flip-Flop to right Flip-Flop for each clock pulse.

EXPERIMENT 13

FINITE STATE MACHINE

AIM:

(1) Design a sequence detector which will detect an overlapping sequence 1010 from an input sequence.

Components and Equipment required:

- IC‟s 7474, 7486, 7473.
- IC trainer kit, connecting wires.

i) . **SEQUENCE DETECTOR**

STEPS:

1. State diagram

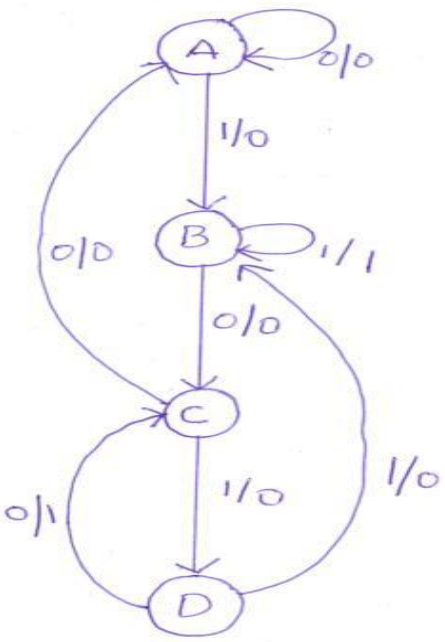


Fig. 13.1 State diagram

2.. State table

STATE	PRESENT STATE A	B	INPUT	NEXT STATE An	Bn	OUTPUT
0	0	0	0	0	0	0
1	0	0	1	0	1	0
2	0	1	0	1	1	0
3	0	1	1	0	1	0
6	1	1	0	0	0	0
7	1	1	1	0	0	0
4	1	0	0	1	1	1
5	1	0	1	0	1	0

On solving using JK flipflop and Kmap following logical expressions are obtained.

$J_A = BX'$ $K_A = BX' + B'X$ $J_B = X+A$ $K_B = A$ $Z = AB'X'$

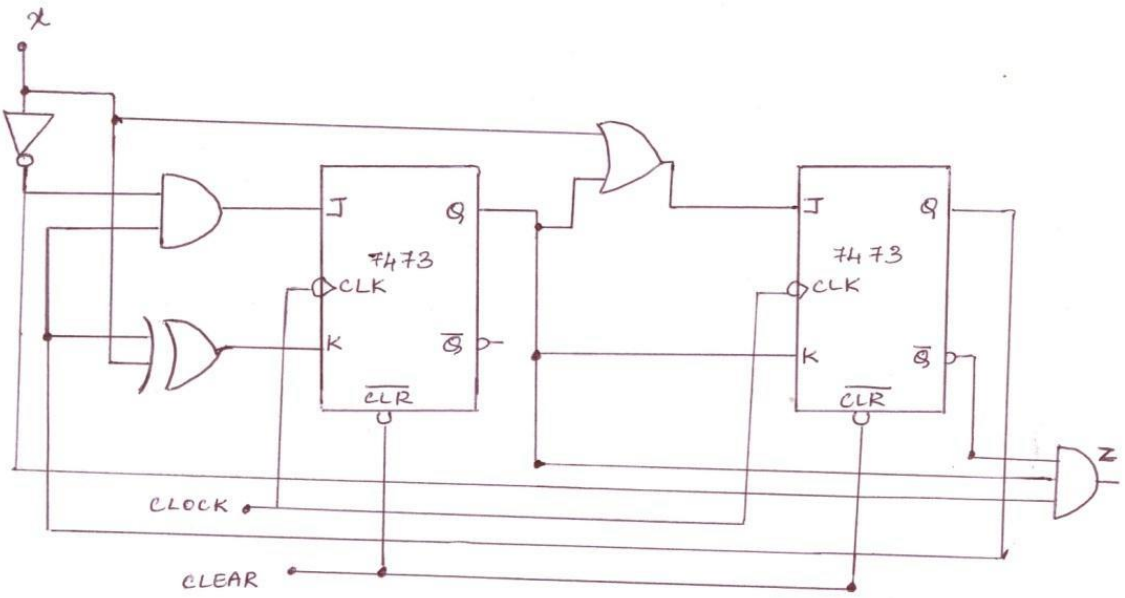


Fig. 13.2

NOTE : Use 7473 IC Dual JK negative edge triggered Flip- flop

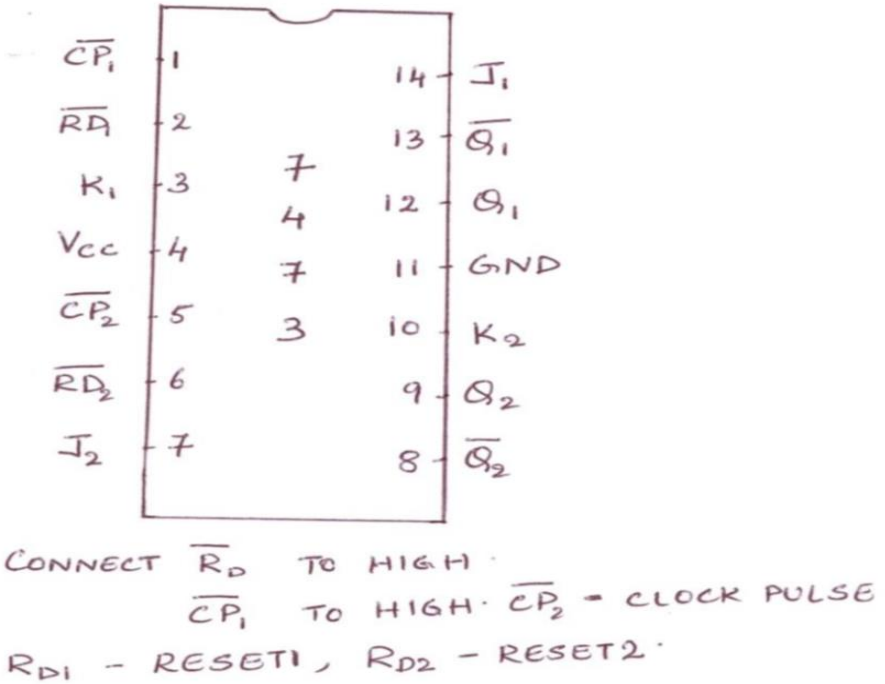


Fig. 13.3 IC diagram of IC 7473

ii) Design an fundamental mode asynchronous sequential circuit with two inputs X and Y and one output Z. Whenever Y is '1', input X is transferred to Z. Whenever Y is '0', output does not change for any change in X.

Components and Equipments Required:

- IC 7408,IC 7404, IC 7432.
- IC Trainer kit, connecting wires

State Table:

PS	NS (XY)				OUT PUT (Z)			
	00	01	10	11	00	01	10	11
A	A	B	--	C	0	--	--	--
B	A	B	D	--	--	0	--	--
C	A	--	D	C	--	--	--	0
D	--	B	D	E	--	--	1	--
E	F	--	D	E	--	--	--	1
F	F	B	--	E	1	--	--	--

Using state table reduction technique the following Table is obtained:

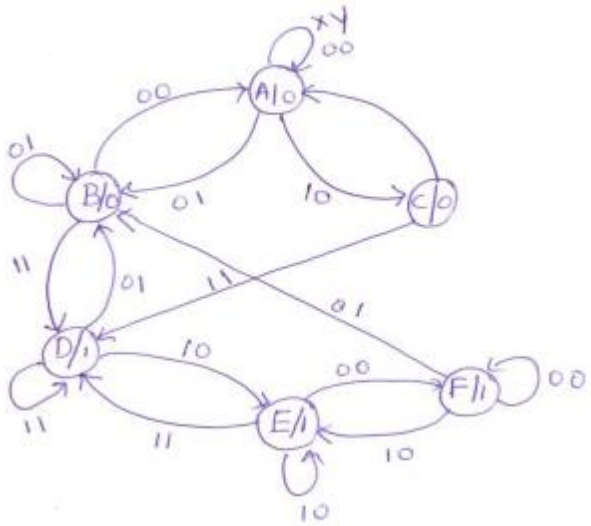
PS	NS				O/P			
	00	01	10	11	00	01	10	11
S0	S0	S0	S1	S0	0	0	--	0
S1	S1	S1	S0	S1	1	--	1	1

Let S0=0, S1=1

PS	NS				O/P			
	00	01	10	11	00	01	10	11
0	0	0	1	0	0	0	--	0
1	1	1	0	1	1	--	1	1

0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	--
1	0	0	1	1
1	0	1	0	--
1	1	0	1	1
1	1	1	1	1

After simplification the following logical expressions are obtained:
F+ =XY +FY‟ and Z=F
State Diagram: Logic Diagram:



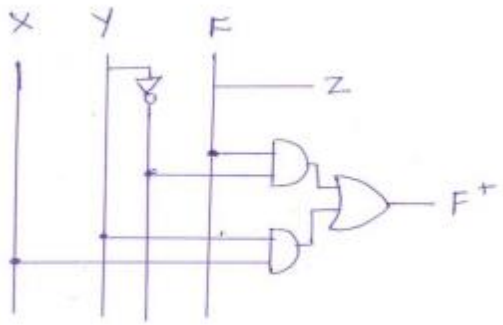


Fig. 13.4 **Fig. 13.5**

Exercises:

(1) Generate the sequence 0-1-2-5-6-7-3-0 using shift register (74LS194).

(2)Design and test a circuit that checks for the sequence 011 continuously in a data sequence using DFF.